

Camera Model + Geometric Transformation

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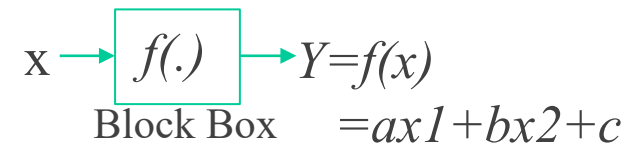
Major Issues

1. Camera Model:

- 1) Intrinsic + Extrinsic Parameters
- 2) Homogenous Matrix

2. Geometric Transformation:

- 1) Affine Transformation
- 2) Perspective Transformation



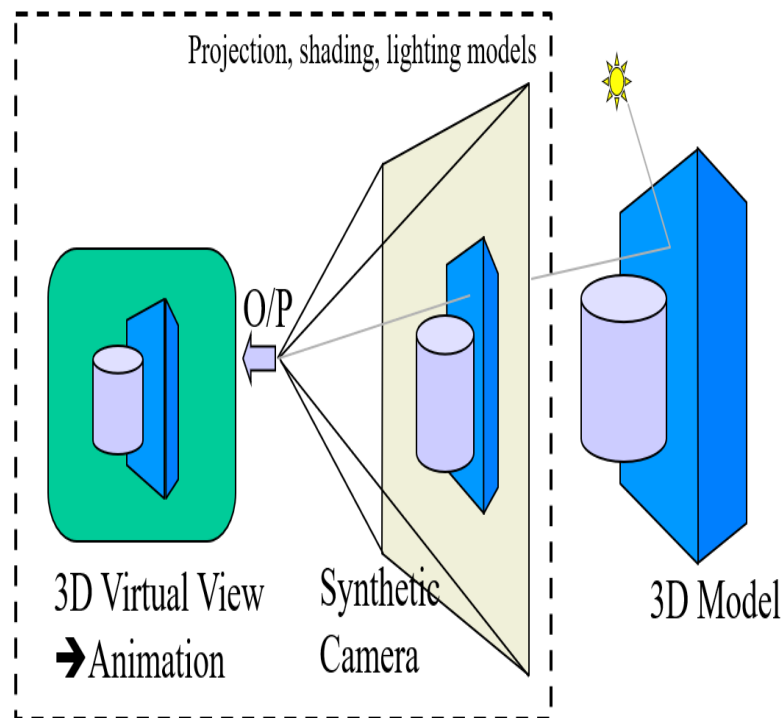
3. Why do we want to know Camera Modeling and Camera Calibration?

➔ Understand the relation btw 2D and 3D

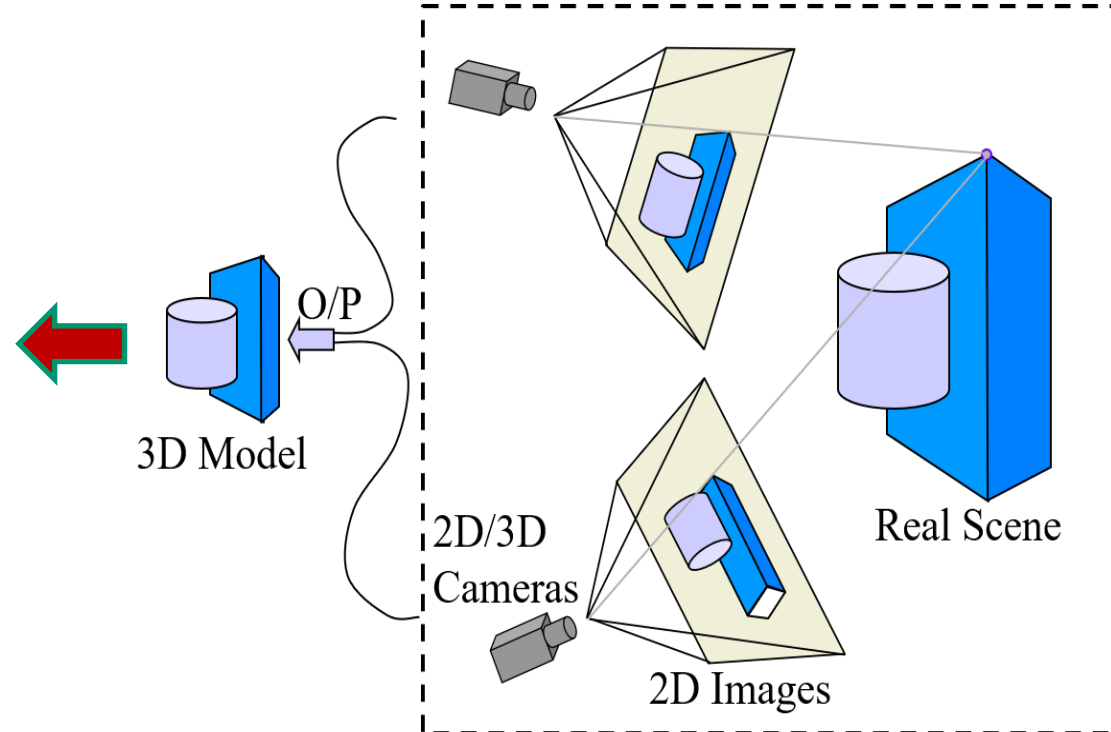
1.0 CV Vs. CG: CV Sensor - Camera

Facebook Metaverse: AR + VR

2) Virtual Reality VR with Computer Graphics



1) AR with Computer Vision



1) AR: Occurs right now

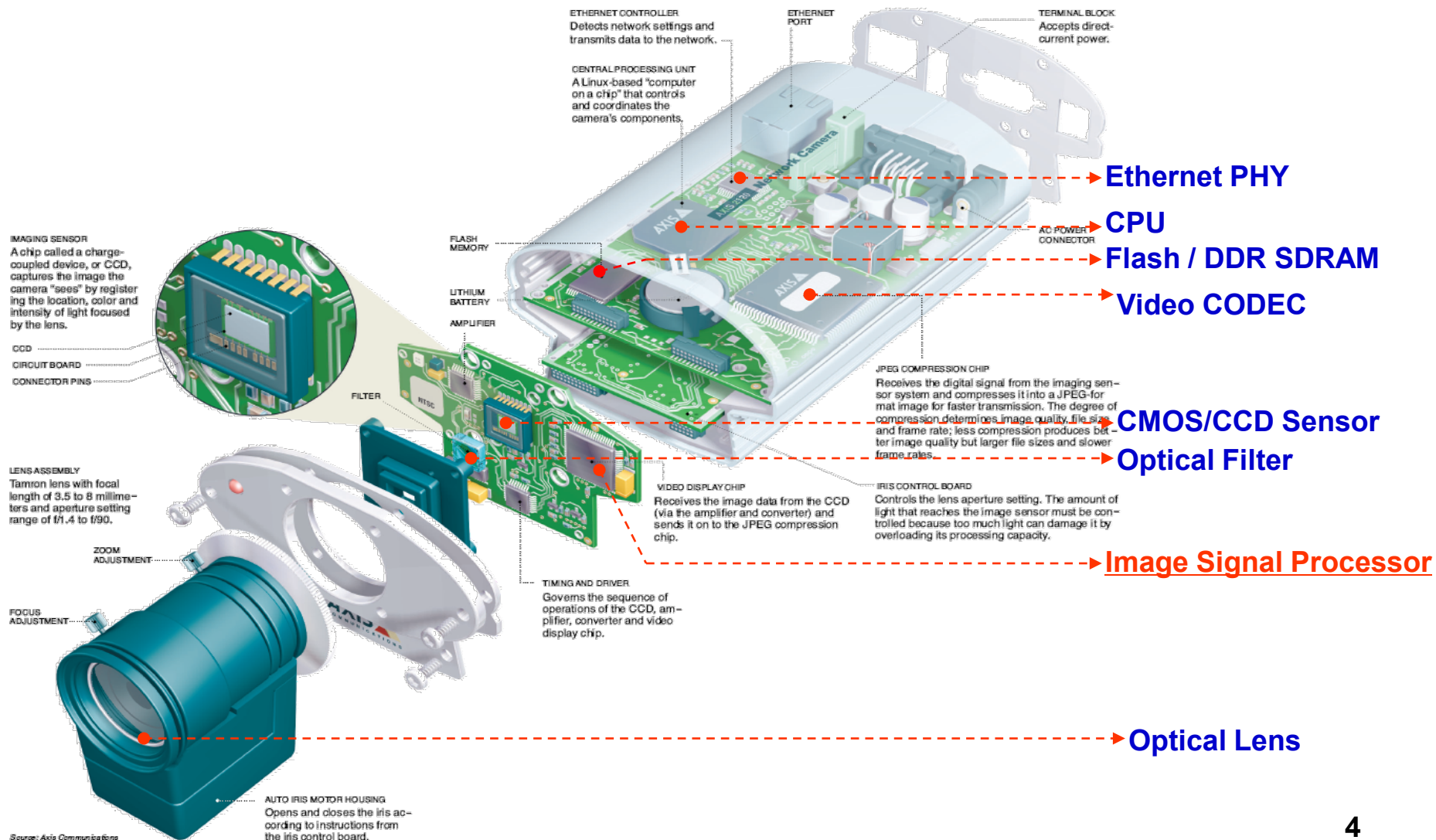
2) VR: Still has a long way to go, 穿戴不方便, 頭暈眩, 但特殊領域可以-內容是什麼?

Tesla Self-Driving Cars: Autopilot 用純視覺, 放棄 LiDAR, 不對吧?

- Wrong Decision-Metaverse at FaceBook and Self-Driving Cars at Tesla
(元宇宙)

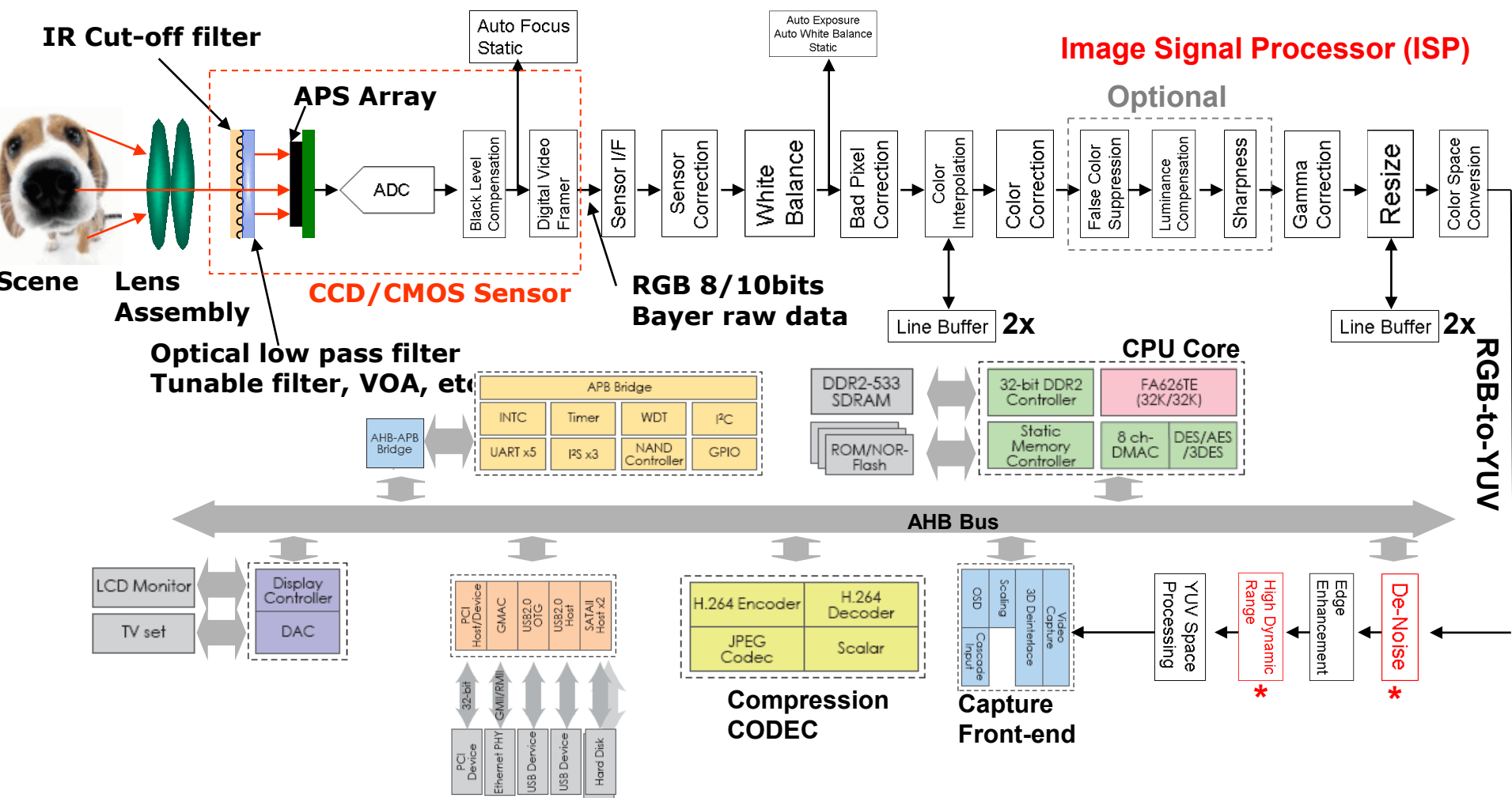
1.1 IP Cam: Image and Signal Processing (ISP) (1/2)

■ IP CAM Block Diagram:



1.1 IP Cam: Image and Signal Processing (ISP) (2/2)

■ IP CAM Function Block Diagram:

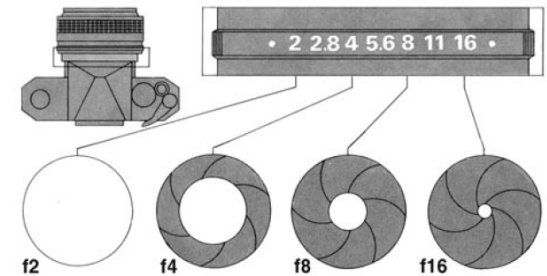
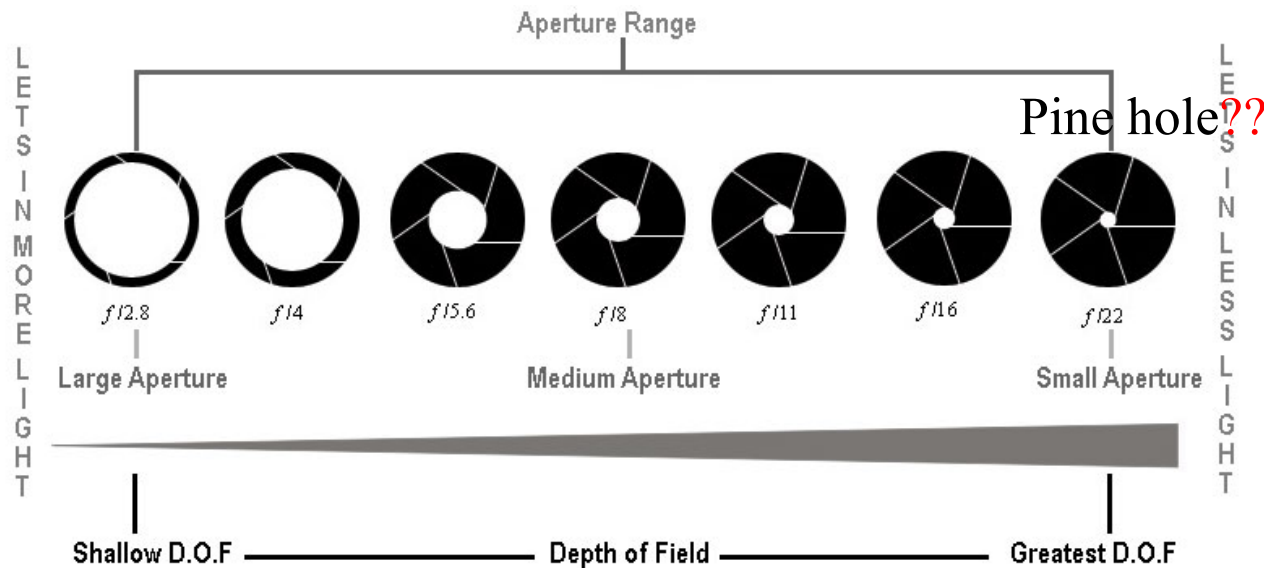


1.1.1 Lens

❑ Lens

- Focusing Control Vs. Focal Length f
- Shutter Speed 快門速度
- Aperture/Iris 光圈大小

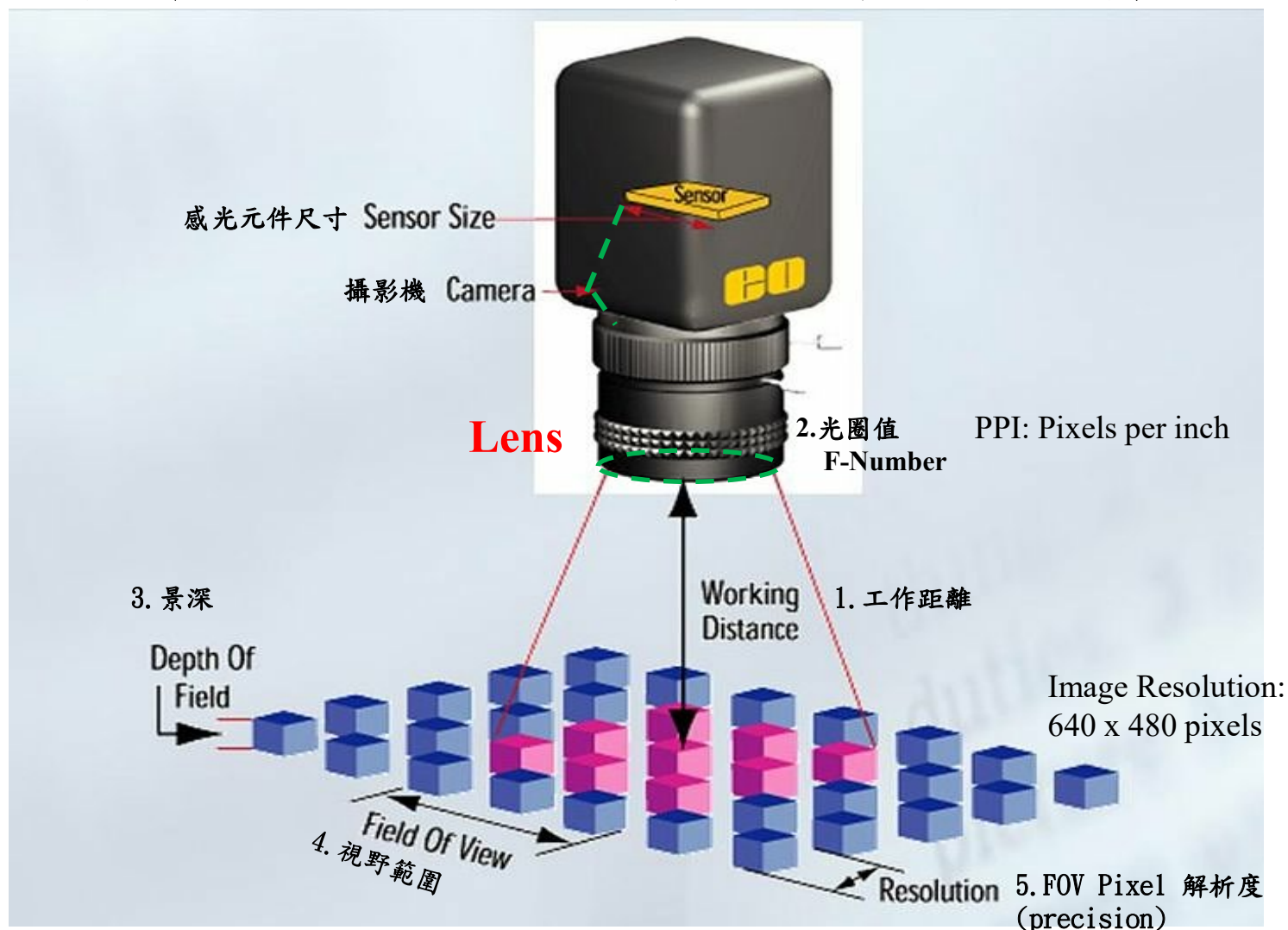
❑ ~~Film~~



1.1.1 鏡頭參數

● 攝影機與鏡頭組合的基本名詞解釋：

鏡頭參數有5個：1.工作距離 2.光圈值 3.景深 4.視野範圍 5.圖形分辨率

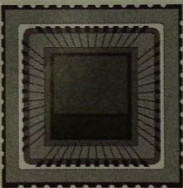


1.1.2 Sensor Size (1/2)

CCD VS. CMOS SENSORS



CCD (Charge Coupled Device) and CMOS (Complimentary Metal Oxide Semiconductor) are different sensor technologies for converting light into electronic signals. In a CCD, each pixel's charge is converted to voltage, buffered, and transferred through a single node as an analog signal. In a CMOS sensor, the charge-to-voltage conversion is done at the pixel level, yielding a less uniform output.

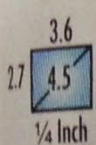
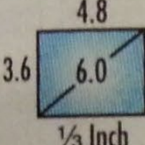
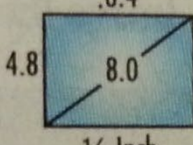
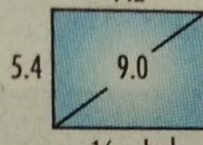
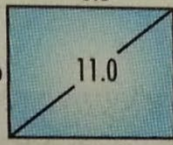
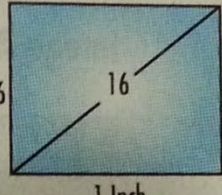
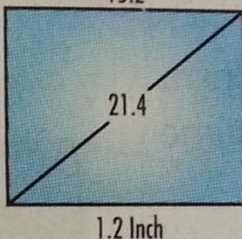


The nonuniformity is especially evident in low light environments, such as applications at high speed or high magnifications. CMOS Sensors should be incorporated into applications requiring low power consumption or into space-constrained applications, whereas CCD's are recommended for applications requiring superior image quality.

CAMERA COMPARISON

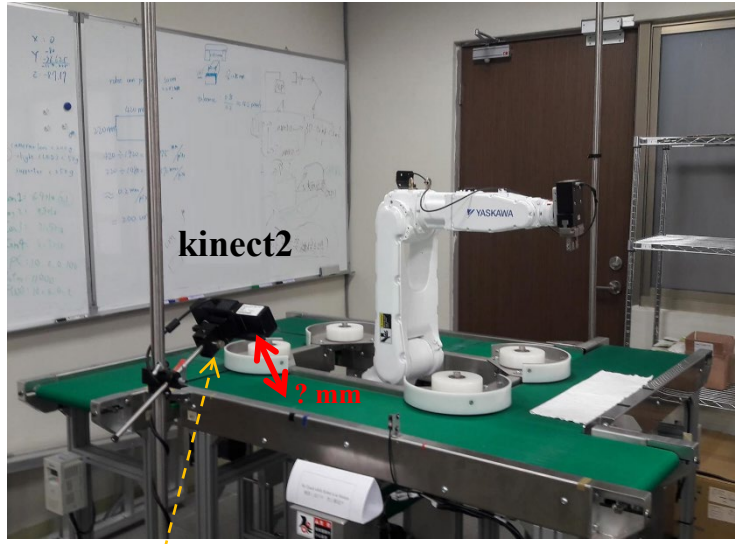
Sensor Size

- Determines System Field Of View (FOV)
- Determines Required Primary Magnification (PMAG)
- Most Have a 4:3 (H:V) Dimensional Aspect Ratio

						
Units: mm						

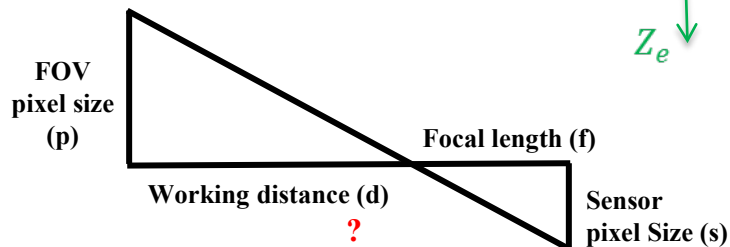
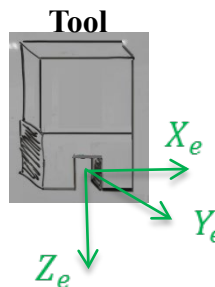
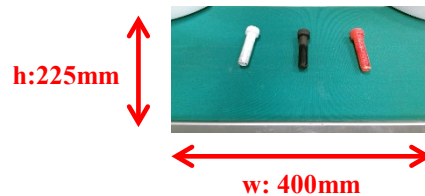
1.1.2 Sensor Size with FOV Precision (2/2)

- Decide precision and camera position:



Move kinect2 outside of the conveyor to avoid occlusion by arm.

Image **I** (1920*1080 pixels) :



Opened Gripper (Max gripper width) 18.2mm

Target (nail width) 16.5mm

Kinect2 Camera Spec.	w	H
Image Resolution	1920 pixels	1080 pixels
Focal Length[1]	3.29mm	
Sensor Size[1]	6.00 mm	3.38 mm
Pixel Size Precision[1]	3.13 $\mu\text{m}/\text{pixel}$	3.13 $\mu\text{m}/\text{pixel}$
Field of View (FOV)	400 mm	225 mm
Pixel size (FOV) ?	208.3 $\mu\text{m}/\text{pixel}$	208.3($\mu\text{m}/\text{pixel}$)

$$\frac{p \text{ (}\mu\text{m/pixel)}}{? d \text{ (mm)}} = \frac{s \text{ (}\mu\text{m/pixel)}}{f \text{ (mm)}}$$

$$\text{width: } P = \frac{\text{FOV}}{\text{Resolution}} = \frac{400,000}{1920} = 208.3(\mu\text{m/pixel})$$

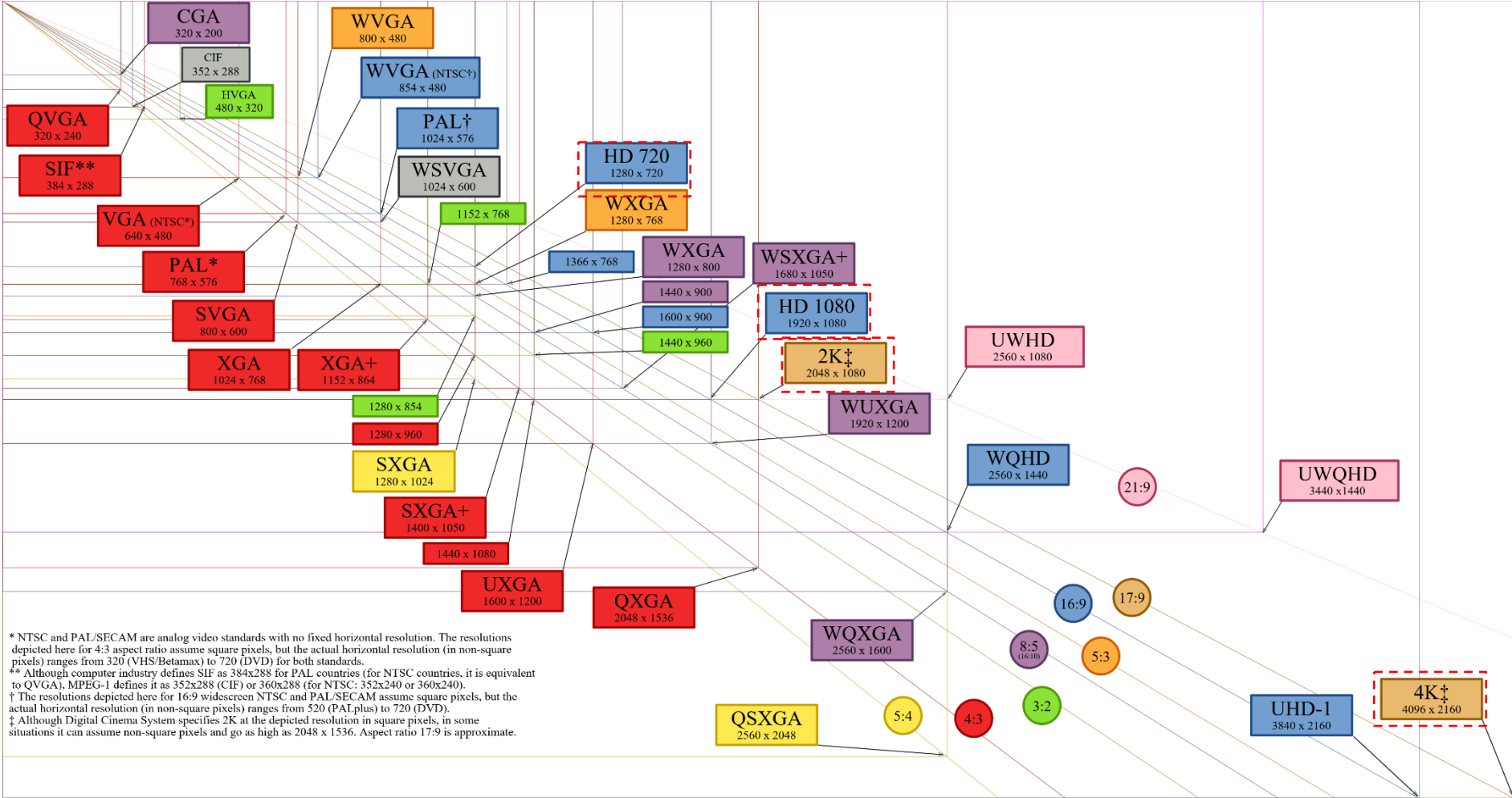
$$\frac{225,000}{1080} = 208.3(\mu\text{m/pixel})$$

$$\frac{208.3}{d} = \frac{3.13}{3.291} \Rightarrow d = 219(\text{mm}) = 21.9(\text{cm})$$

View-based gripper one side tolerance:

$$\frac{(18.2\text{mm}-16.5\text{mm})/2}{0.2083 \text{ mm}} = \frac{0.85\text{mm}}{0.2083 \text{ mm}} = 4.1 \text{ pixels}$$

1.1.3 Display Resolution (1/2)

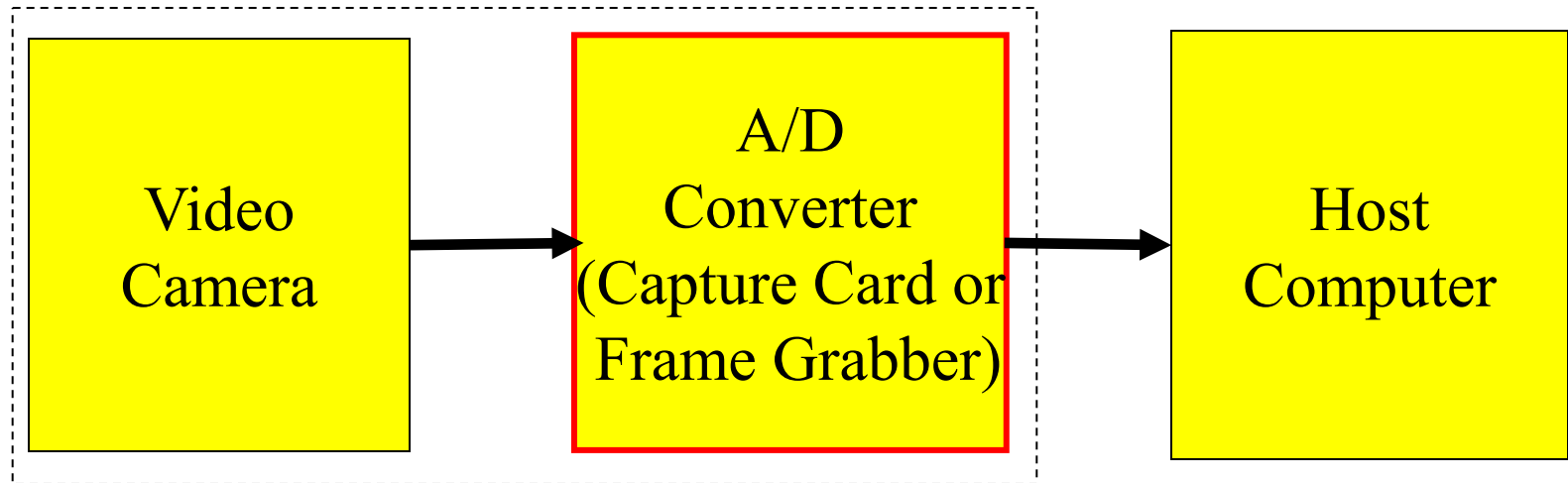


1.1.3 Television Resolutions (2/2)

- **Standard-definition television ([SDTV](#)):**
 - [480i](#) ([NTSC-compatible](#) digital standard employing two [interlaced](#) fields of 243 lines each)
 - [576i](#) ([PAL-compatible](#) digital standard employing two interlaced fields of 288 lines each)
- **Enhanced-definition television ([EDTV](#)):**
 - [480p](#) (720×480 [progressive scan](#))
 - [576p](#) (720×576 progressive scan)
- **High-definition television ([HDTV](#)):**
 - [HD](#) (1280×720 progressive scan)
 - [Full HDi](#) (1920×1080 split into two interlaced fields of 540 lines)
 - [Full HD](#) (1920×1080 progressive scan)
- **Ultra-high-definition television ([UHDTV](#)):**
 - [4K UHD](#) (3840×2160 progressive scan)
 - [True 4K](#) (4096×2160)
 - [8K UHD](#) (7680×4320 progressive scan)
 - [True 8K](#) (8192×4320)

1.2 Digital Imaging Systems

- ❑ **CCD or CMOS imaging array**
 - When light falls on the cells in these arrays, a charge accumulates which is proportional to the incident light energy
- ❑ **Analog to digital conversion unit**
- ❑ **Host Computer**



1.2 Image Color

- ☐ RGB
 - ☐ **HSV** $\Leftrightarrow$ **RGB**, (check powerpoint character color map)
 - ☐ YCrCb
 - ☐ CIE
 - ☐ CMYK
 - ☐ CIE LAB
 - ☐ ...
-
- ☐ **H.264** $\Leftrightarrow$ **YUV** $\Leftrightarrow$ **RGB**
 - ☐ **H.265**: 3D

2.0 Pinhole Camera Model

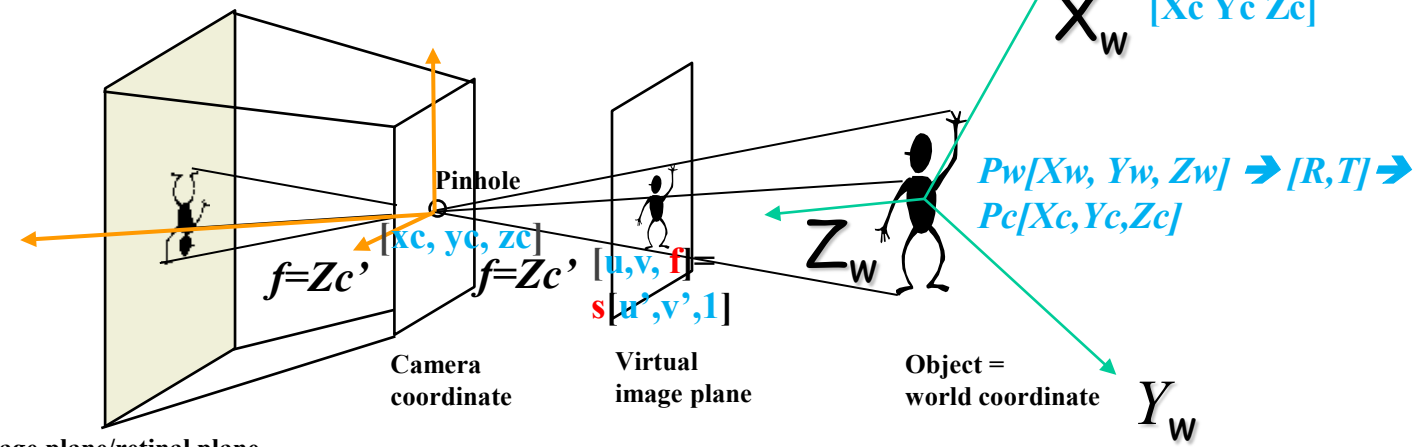
Pinhole Camera Model

➤ Camera Parameters:

- 1) 5 Intrinsic parameters
- 2) 5 Lens distortion parameters
- 3) 12 Extrinsic parameters: $R(3 \times 3) + T(1 \times 3)$

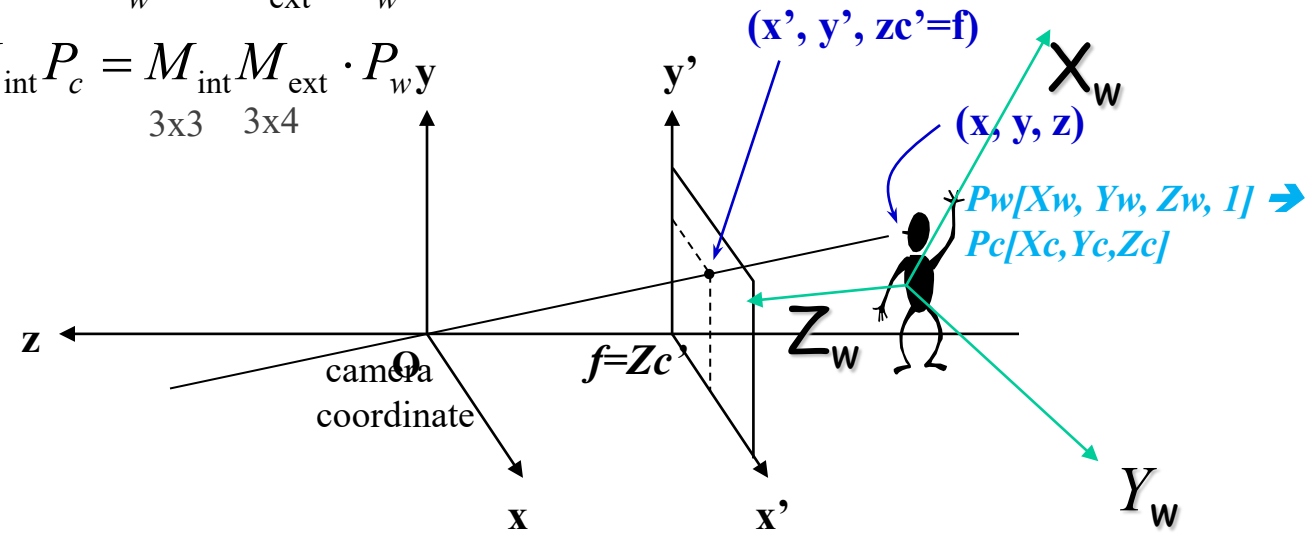
2.1 Pine Hole Camera Model: Perspective Projection (1/4)

3D World to 3D Camera to 2D Image Coordinate



W to C:
$$P_c = R \cdot T \cdot P_w = M_{ext} \cdot P_w$$

C to I:
$$p = M_{int} P_c = M_{int} M_{ext} \cdot P_w$$

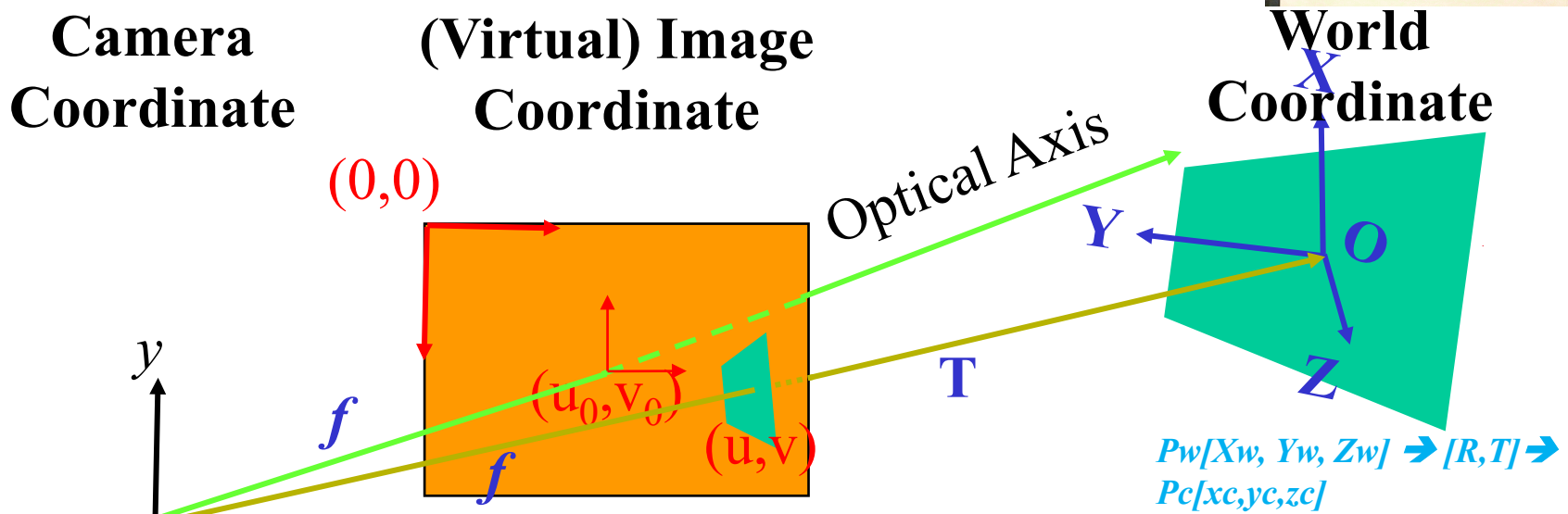
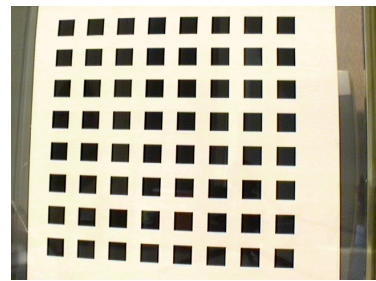


2.1 World to Camera to Image Coordinates (2/4)

3D 3D 2D

OpenCV Ch. 11

- Camera Calibration by Chessboard or Chessboard pattern
 → to estimate 12 Extrinsic and 5 Intrinsic+5 Distortion parameters



$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} R_{11} & R_{21} & R_{31} \\ R_{12} & R_{22} & R_{32} \\ R_{13} & R_{23} & R_{33} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix}$$

Extrinsic Parameters: 6 DOF

I

□ Affine Transformation

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Intrinsic Parameters = Projection matrix

2.1 Camera Parameters: Intrinsic + Extrinsic Parameters (3/4)

$$\begin{aligned}
 \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} &= \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \text{Orthonormal unit vector} \\ \begin{bmatrix} R_{11} & R_{21} & R_{31} \\ R_{12} & R_{22} & R_{32} \\ R_{13} & R_{23} & R_{33} \end{bmatrix} & \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \\
 &\quad \begin{matrix} \text{Scale Factor: } \lambda & \text{[Xc Yc Zc]} & \text{Normalize to unit vector} \end{matrix}
 \end{aligned}$$

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$s \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{14} \\ h_{41} & h_{42} & h_{43} & 1 \end{bmatrix} \begin{bmatrix} X_n^w \\ Y_n^w \\ Z_n^w \\ 1 \end{bmatrix}$$

Homogenous Matrix

$$\begin{aligned}
 u &= -f k_u (x/z) + \gamma (y/z) + u_0 \\
 v &= -f k_v (y/z) + v_0
 \end{aligned}$$

Equation

Intrinsic Parameters:

- **Scale Factor:** $\alpha = -f k_u$, f : focal length
- **Scale Factor:** $\beta = -f k_v$
- Aspect Ratio** = $\alpha/\beta = k_u/k_v \approx 1$
- **Skew Factor:** γ
- **Principal Point:** (u_0, v_0)

Extrinsic Parameters:

- **Rotation:** $\mathbf{R}$
- **Translation:** $\mathbf{T}$

OpenCV Ch. 11

1 pattern at RAM

$$\min \sum_{n=1}^N (p_n^u - p_n^v) \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix}^s = \begin{bmatrix} h11 & h12 & h13 & h14 \\ h41 & h42 & h43 & 1 \end{bmatrix} \begin{bmatrix} X_n^w \\ Y_n^w \\ Z_n^w \\ 1 \end{bmatrix}$$

- (1) Radial (barrel) distortion:**

$$u_n^u = u_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

$$v_n^u = v_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

- (2) **Tangential distortion:**

$$u_n^u = u_n + [2p_1 v_n + p_2(r^2 + 2u_n^2)]$$

$$v_n^u = v_n + [p_1(r^2 + 2v_n^2) + 2p_2u_n] \quad \text{where } r^2 = u_n^2 + v_n^2$$

- 2) Perspective Projection: From 3D camera coordinate projects to 2D image coordinate**

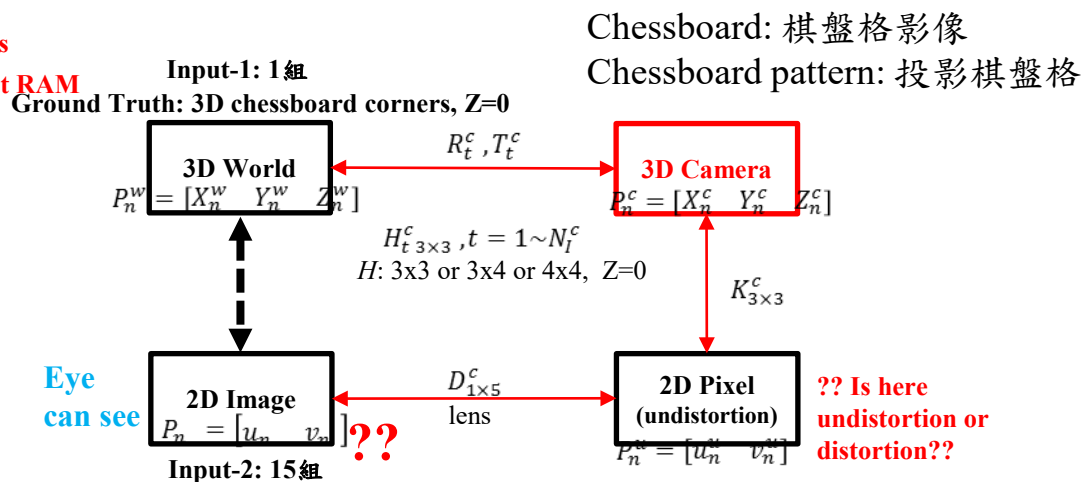
$$s \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & \gamma = 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_n^c \\ Y_n^c \\ Z_n^c = 0 \\ 1 \end{bmatrix}$$

- ### 3) Affine: From 3D world coordinate to 3D camera coordinate

$$S \begin{bmatrix} X_n^c \\ Y_n^c \\ Z_n^c = 0 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_n^{w'} \\ Y_n^{w'} \\ Z_n^{w'} = 0 \end{bmatrix}$$

- #### 4) From 3D world coordinate to 2D coordinate

$$s \begin{bmatrix} u_n \\ v_n \\ I \end{bmatrix} = \begin{bmatrix} f_x & \gamma = 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_n^{w'} \\ Y_n^{w'} \\ Z_n^{w'} = 0 \\ 1 \end{bmatrix}$$



- Camera Calibration by Chessboard or Chessboard pattern
→ to estimate 12 Extrinsic and 5 Intrinsic+5 Distortion parameters

Rodrigues Transform: 1x3

Rotation
Matrix:
3x3 Matrix

Eular Angle: 1x3

- ## ► Camera Parameters:

- 1) 12 Extrinsic parameters:

$$R(3 \times 3) + T(1 \times 3)$$

- ## 2) 5 Intrinsic parameters

- ### 3) 5 Lens distortion parameters

2.1.1 Camera Parameters: Intrinsic + Extrinsic Parameters

- ❑ **Intrinsic parameters (Camera Coordinate $\Leftrightarrow$ Image Coordinate)**
 - Do not depend on the camera location
 - » Focal length f , CCD dimensions (u_0, v_0) , lens distortion
 - ❑ **Extrinsic parameters (World Coordinate $\Leftrightarrow$ Camera Coordinate)**
 - Depend on the camera location
 - » Translation, and Rotation parameters
-

- ❑ **Intrinsic parameters: Projection matrix**
 - Is calculated via **camera calibration**
 - Most important intrinsic parameter is **focal length f** .
 - » **3D-to-2D projection**: From 3D world/camera coordinate projects to 2D image coordinate.
 - » **2D-to-3D back-projection/reconstruction**: From 2D image coordinate back-projects to 3D world coordinate.
- ❑ **Extrinsic parameters: Affine transformation**
 - Geometric transformation
 - » **3D** world/camera coordinate Vs. **3D** world/camera coordinate
 - » **2D** image coordinate Vs. **2D** image coordinate

2.1.1 Pinhole Camera Properties

- ☐ **Non-linear equations**
- ☐ **Lines project into lines**
- ☐ **Does not preserve angles**
- ☐ **Circles project into ellipses**
- ☐ **Farther objects appear smaller**

2.1.2 1) Radial Distortion

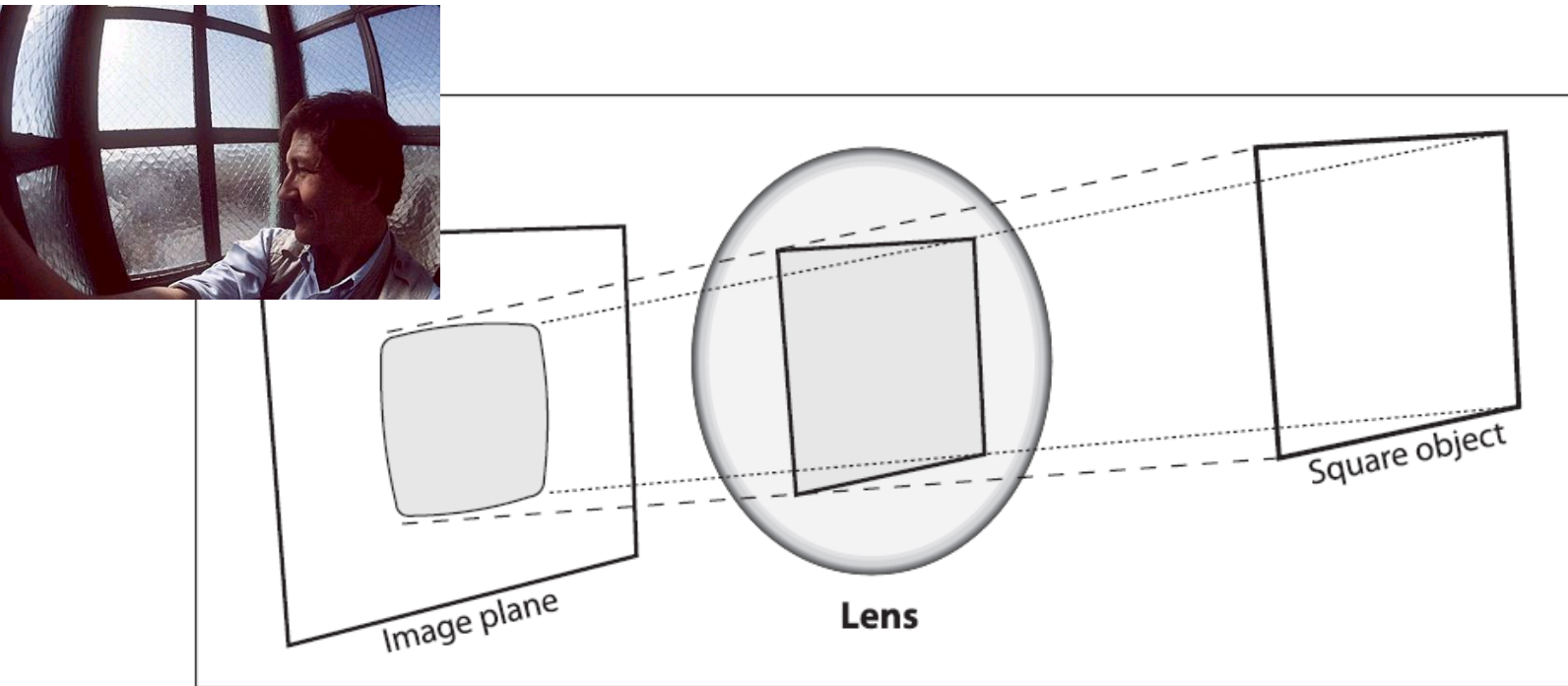


Figure 11-3. Radial distortion: rays farther from the center of a simple lens are bent too much compared to rays that pass closer to the center; thus, the sides of a square appear to bow out on the image plane (this is also known as **barrel distortion**)

$$x_{\text{corrected}} = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) \quad r^2 = x^2 + y^2$$

$$y_{\text{corrected}} = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

Here, (x, y) is the original location (on the imager) of the distorted point and $(x_{\text{corrected}}, y_{\text{corrected}})$ is the new location as a result of the correction. Figure 11-4 shows displacements of a rectangular grid that are due to radial distortion. External points on a front-facing rectangular grid are increasingly displaced inward as the radial distance from the optical center increases.

2.1.2

2) Tangential Distortion

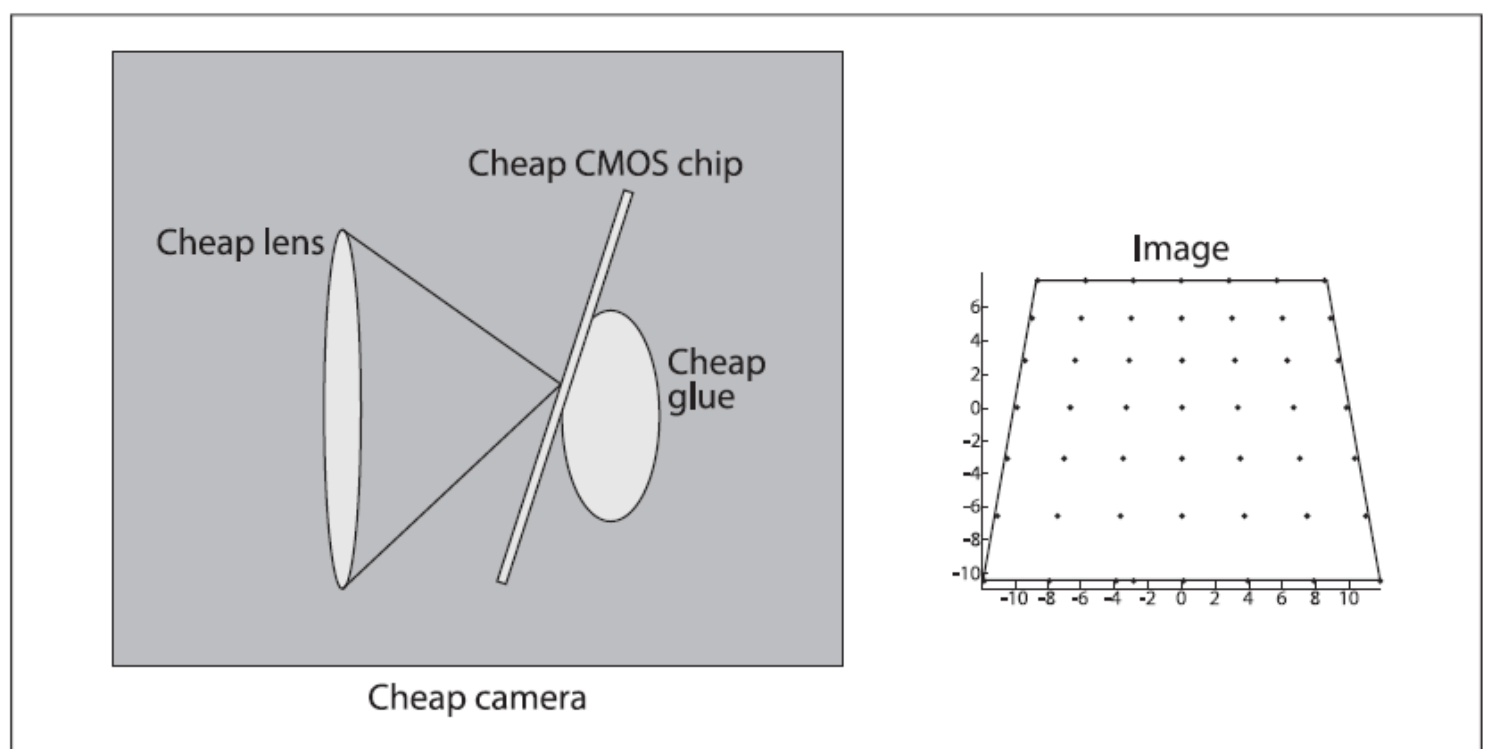


Figure 11-5. Tangential distortion results when the lens is not fully parallel to the image plane; in cheap cameras, this can happen when the imager is glued to the back of the camera (image courtesy of Sebastian Thrun)

The second-largest common distortion is **tangential distortion**. This distortion is due to manufacturing defects resulting from the lens not being exactly parallel to the imaging plane; see Figure 11-5.

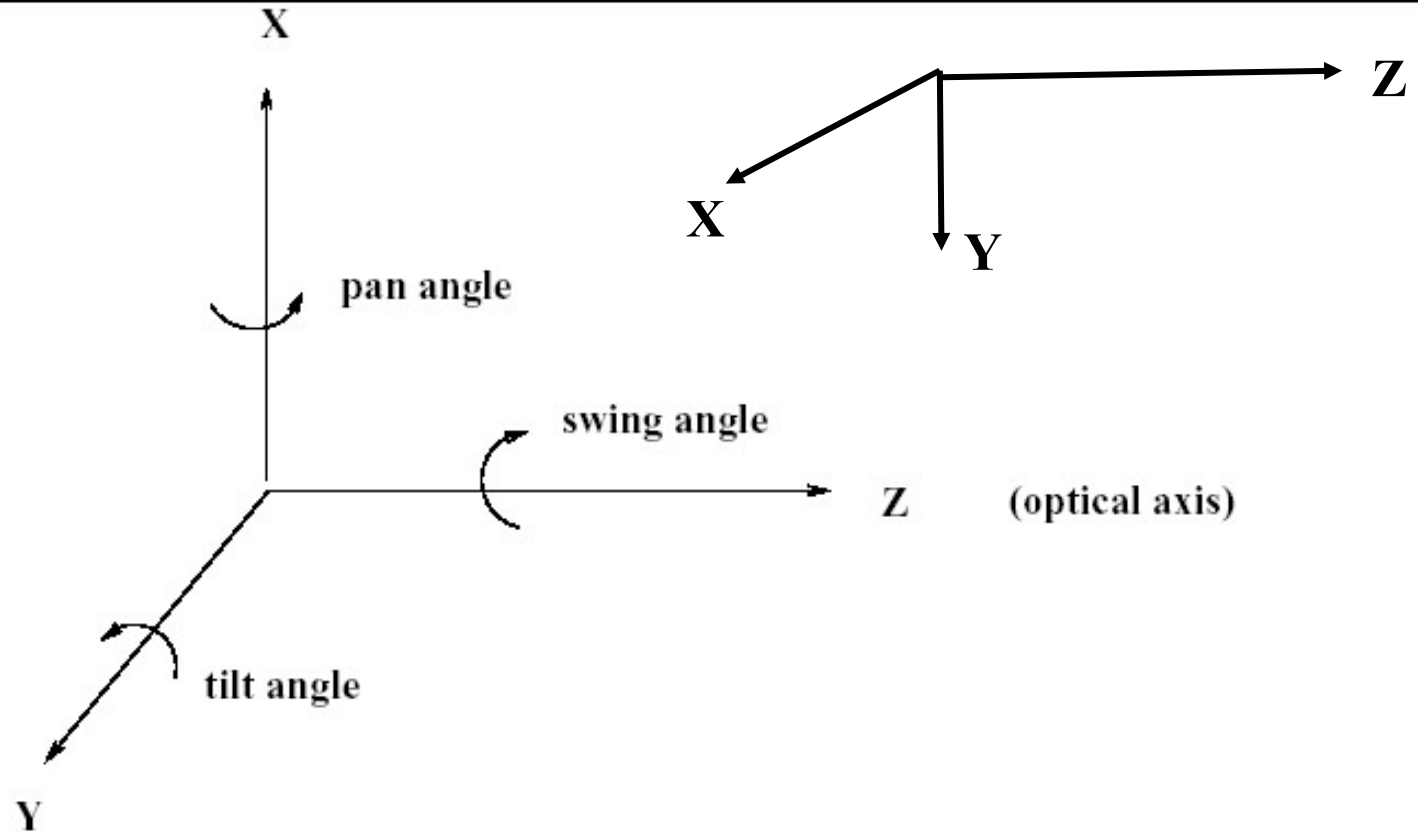
Tangential distortion is minimally characterized by two additional parameters, p_1 and p_2 , such that:^{*}

$$\begin{aligned} x_{\text{corrected}} &= x + [2p_1y + p_2(r^2 + 2x^2)] \\ y_{\text{corrected}} &= y + [p_1(r^2 + 2y^2) + 2p_2x] \end{aligned} \quad r^2 = x^2 + y^2$$

Thus in total there are **five** distortion coefficients that we require. Because all five are necessary in most of the OpenCV routines that use them, they are typically bundled into one **distortion vector**; this is just a 5-by-1 matrix containing k_1 , k_2 , p_1 , p_2 , and k_3

2.1.3

Rotation Matrix Representation: Euler angles



- **In-Plane Rotation/Motion:** Roll (or Swing)
- **Out-of-Plane Rotation Motion:** Pitch (or Tilt) and Yaw (or Pan)

2.1.3

Rotation Matrix Representation: Euler angles

Assume rotation matrix R results from successive Euler rotations of the camera frame around its X axis by ω , its once rotated Y axis by ϕ , and its twice rotated Z axis by κ , then

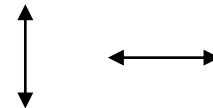
$$R(\omega, \phi, \kappa) = R_X(\omega)R_Y(\phi)R_Z(\kappa)$$

Different
orders have
different results

where ω , ϕ , and κ are often referred to as tilt, pan, and swing angles respectively.



roll



pitch yaw

2.1.3

□ DIP book

$$R_x(\omega) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \omega & \sin \omega \\ 0 & -\sin \omega & \cos \omega \end{pmatrix}$$

Orthonormal unit vector

$$R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & -\sin \phi \\ 0 & 1 & 0 \\ \sin \phi & 0 & \cos \phi \end{pmatrix}$$

$$R_z(\kappa) = \begin{pmatrix} \cos \kappa & \sin \kappa & 0 \\ -\sin \kappa & \cos \kappa & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2.2 3D Reconstruction: Perspective Vs. Orthogonal (Parallel) Vs. Weak Perspective Projection Model

❑ Perspective (pinhole camera model)

- Size varies inversely with distance – looks realistic
- Distance and angles are not preserved (in general)
- Parallel lines do not remain parallel (in general)

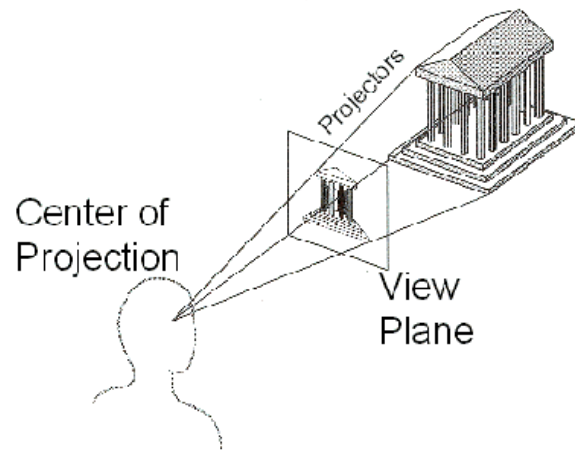
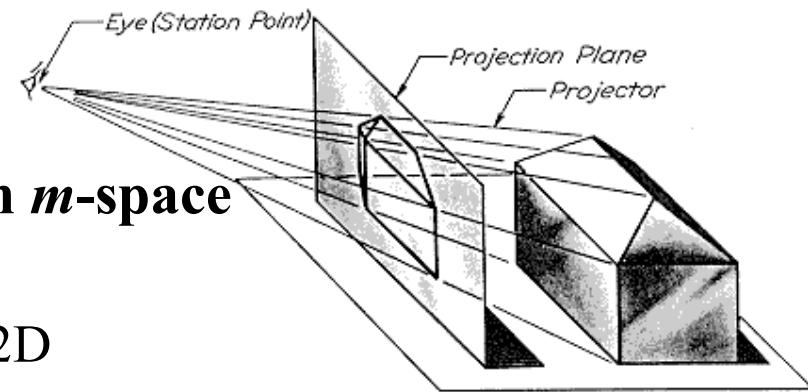
❑ Orthogonal

- Good for exact measurements
- Parallel lines remain parallel
- Angles are not preserved (in general)
- Less realistic looking

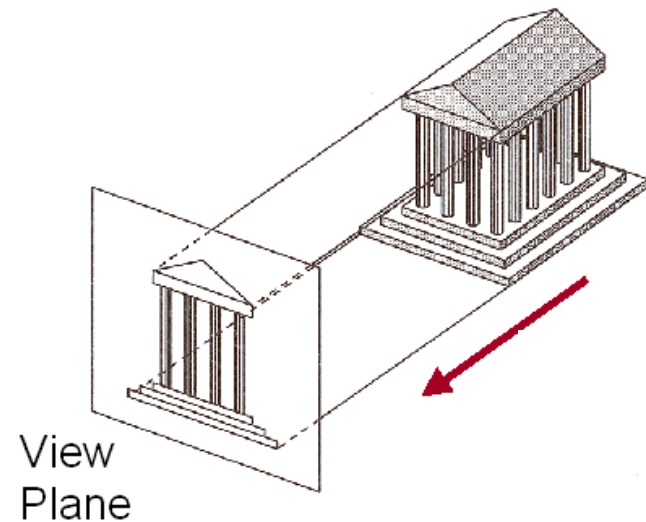
2.2 Projective Geometry

□ **Projection** – a transformation from m -space to n -space ($m > n$)

➤ For vision and graphics, it's 3D to 2D

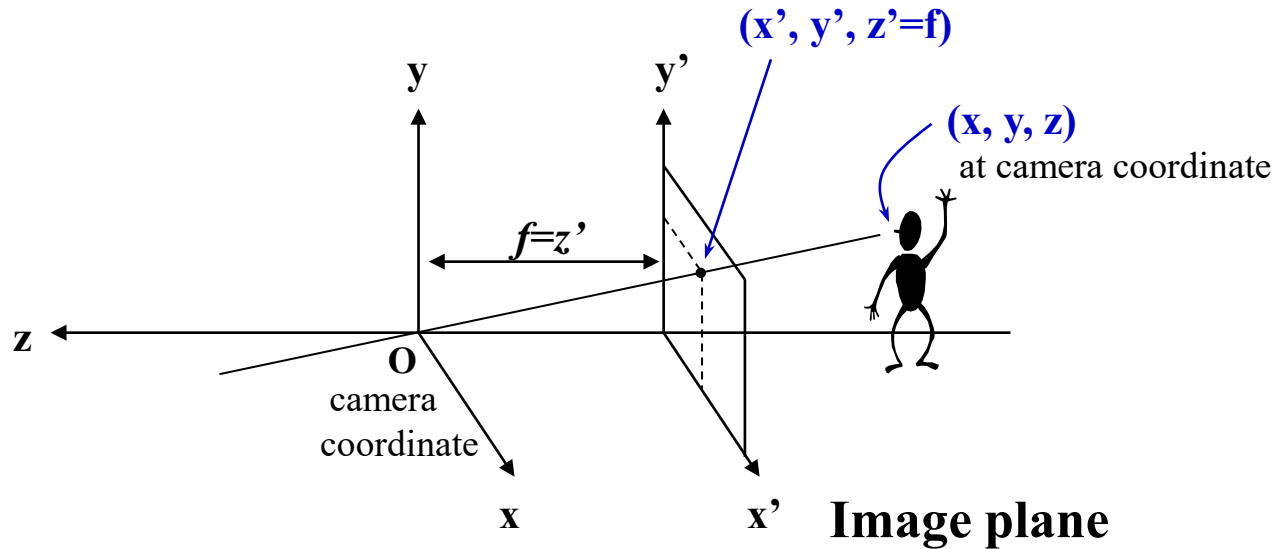


Perspective planar



Orthogonal planar

2.2.1 Perspective Projection Model

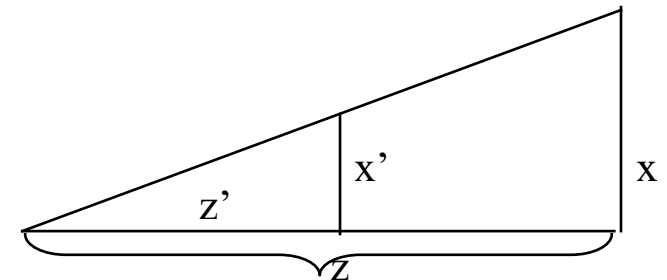


Projection from (x, y, z) to (x', y', z') :

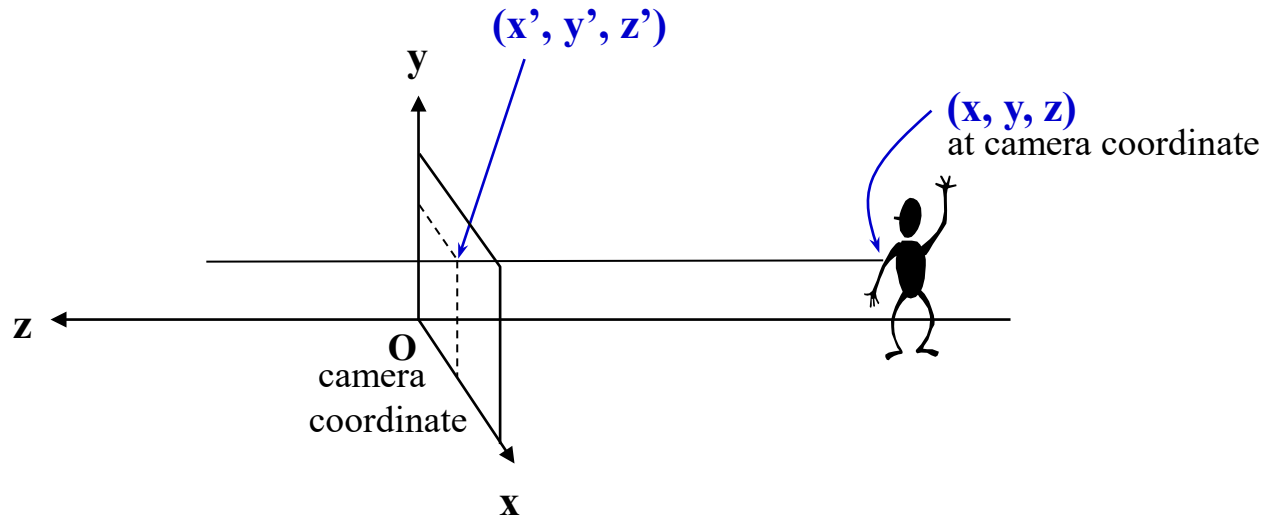
$$x' = z' \frac{x}{z}$$

$$y' = z' \frac{y}{z}$$

$$z' = z' = f$$



2.2.2 Orthogonal/Orthographic Projection Model



Projection from (x, y, z) to (x', y', z') :

$$x' = x$$

$$y' = y$$

$$z' = \text{const}$$

or

$$x' = kx$$

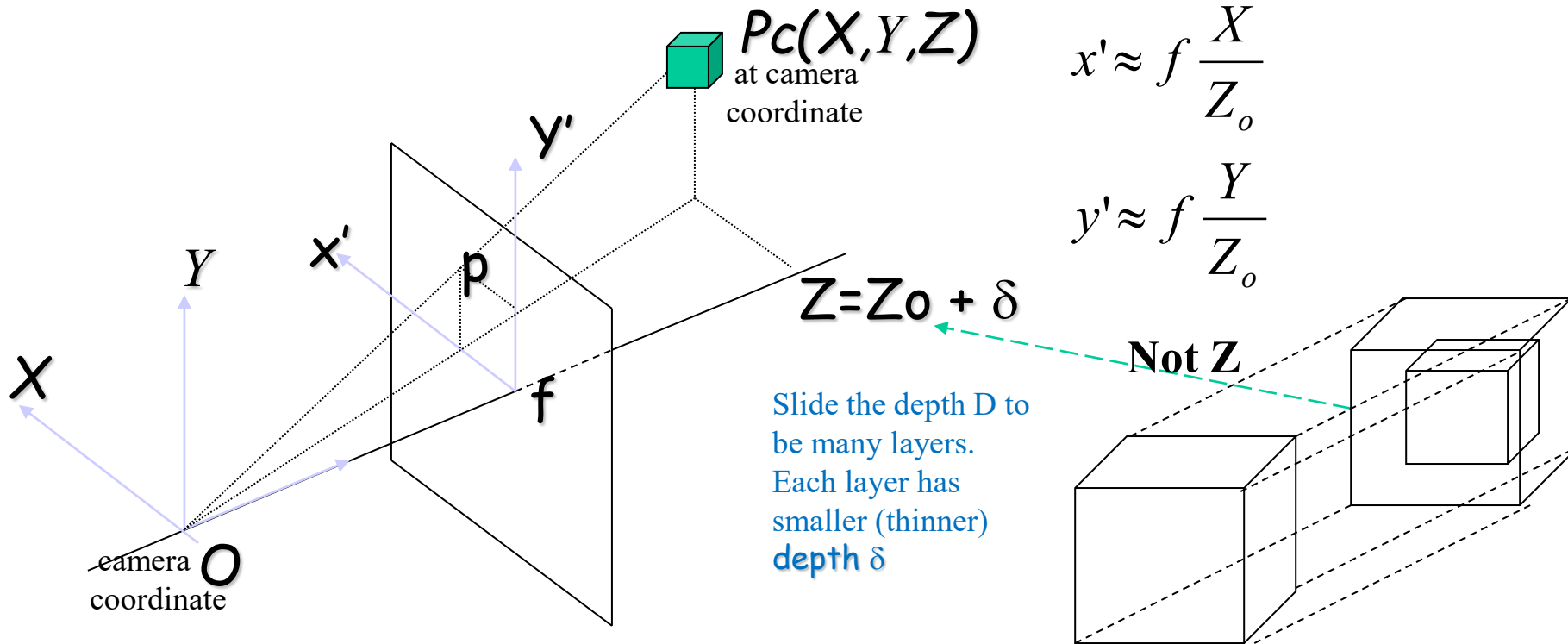
$$y' = ky$$

$$z' = \text{const}$$

2.2.3 Weak Perspective Model

Divide an object to different small layers

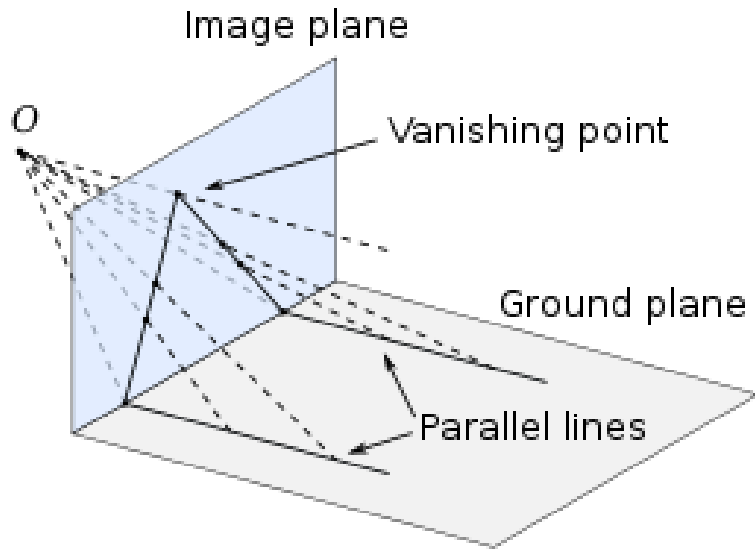
- Appropriate when the depth of field of the relevant objects is small compared to the distance of those objects from the center of projection



- Object depth $\delta \ll$ Camera distance Z_o
- Linear equations !!

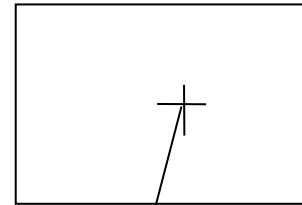
Weak perspective = Orthographic projection + Isotropic Scaling

2.2.4 Vanishing Point (Assist Camera Calibration for 2D to 3D)

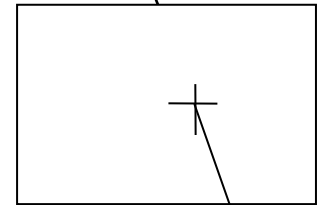


**Infinite point =
Vanishing point**

**Principal point1 !=
center**



**Principal point2 !=
center**



**Principal point1 position at image 1 !=
Principal point2 position at image 2**

2.3 3D World to 3D Camera to 2D Image Coordinate

■ Using homogeneous coordinates:

□ **W to I:**

$$p(x,y) = M \cdot P_w(X_w,Y_w,Z_w)$$

M is 3x4

M has 6 dof (3 R + 3 T)

(assuming f is known)

11 or 12 unknown para. of perspective project matrix, so it needs how many corresponding points btw 3D and 2D? At least 4 as a plane

□ Each image point $p(x, y)$ must satisfy:

1) Linear: Homogenous matrix M , $b=\dot{M}x$

$$\begin{bmatrix} x' \\ y' \\ ?s' \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{31} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

s' as focal length f as scaling factor

2) Non-linear equations

$$\begin{cases} x = \frac{x'}{s'} = \frac{m_{11}X + m_{12}Y + m_{13}Z + m_{14}}{m_{31}X + m_{32}Y + m_{33}Z + m_{34}} \\ y = \frac{y'}{s'} = \frac{m_{21}X + m_{22}Y + m_{23}Z + m_{24}}{m_{31}X + m_{32}Y + m_{33}Z + m_{34}} \end{cases}$$

Linear Combination $y=w1*x1+w2*x2+b =$

$$f(x) = a_0 + \sum_{n=1} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$0 = m_{11}X + m_{12}Y + m_{13}Z + m_{14} - m_{31}Xx - m_{32}Yx - m_{33}Zx - m_{34}x$$

$$0 = m_{21}X + m_{22}Y + m_{23}Z + m_{24} - m_{31}Xy - m_{32}Yy - m_{33}Zy - m_{34}y$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{21} & R_{31} & T_1 \\ R_{12} & R_{22} & R_{32} & T_2 \\ R_{13} & R_{23} & R_{33} & T_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Scale Factor: λ ? Normalize to unit vector

[Xc Yc Zc]

2.3 3D World to 3D Camera to 2D Image Coordinate

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{21} & R_{31} & T_1 \\ R_{12} & R_{22} & R_{32} & T_2 \\ R_{13} & R_{23} & R_{33} & T_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Scale Factor: λ ? Normalize to unit vector ? [Xc Yc Zc]

• Extrinsic parameters (R, T):

W to C: $P_c = R \cdot T \cdot P_w = M_{\text{ext}} \cdot P_w$

3x1 3x4 3x4 4x1

? ?

• Intrinsic parameter (f):

C to I: $p = M_{\text{int}} P_c = M_{\text{int}} M_{\text{ext}} \cdot P_w$

3x1 3x3 3x4 3x1

? ? ?

W to I:

$$p = M \cdot P_w$$

?

M is 3x4

M has 6 dof (3 R + 3 T)
(assuming f is known)

3x3 Rotation

Matrix

[R_x, R_y, R_z]

1x3 Translation

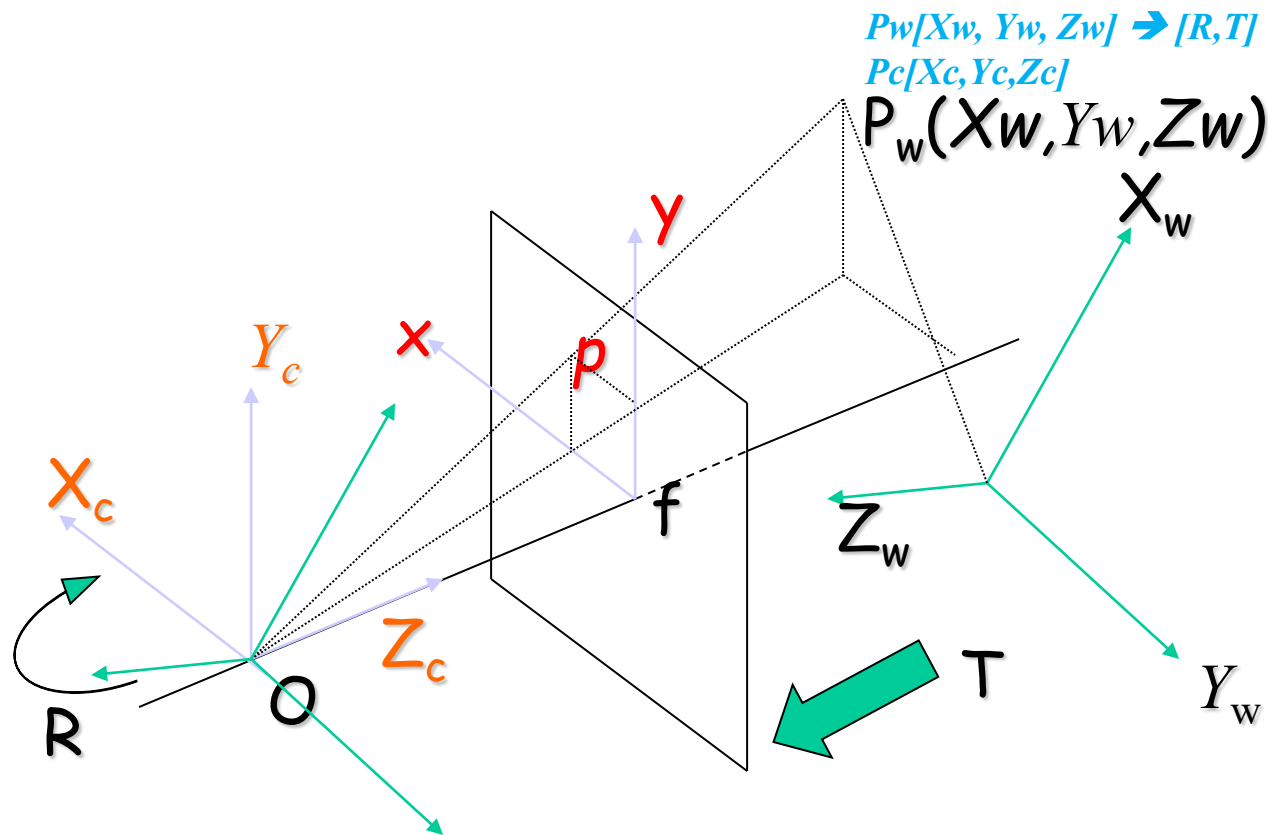
Vector

[T_x, T_y, T_z]

2.3 3D Camera to 2D Image Coordinate

$$\lambda \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{1} \end{bmatrix} = \begin{pmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} R_{11} & R_{21} & R_{31} & T_1 \\ R_{12} & R_{22} & R_{32} & T_2 \\ R_{13} & R_{23} & R_{33} & T_3 \end{pmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Scale Factor: λ $\begin{matrix} \uparrow & \uparrow & \uparrow \\ \text{Normalize to unit vector} \end{matrix}$



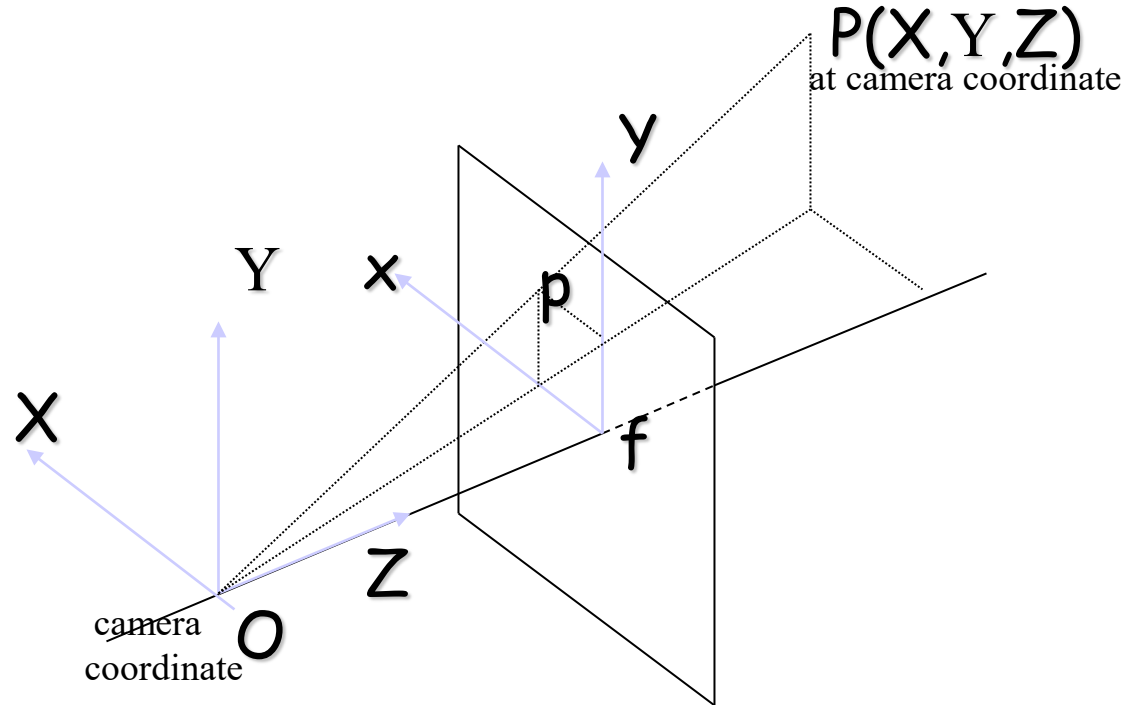
W to C: $P_c = R \cdot T \cdot P_w = M_{\text{ext}} \cdot P_w$

C to I: $p = M_{\text{int}} P_c = M_{\text{int}} M_{\text{ext}} \cdot P_w$

2.3 3D Camera to 2D Image Coordinate: X-Y-Z Camera Coordinate

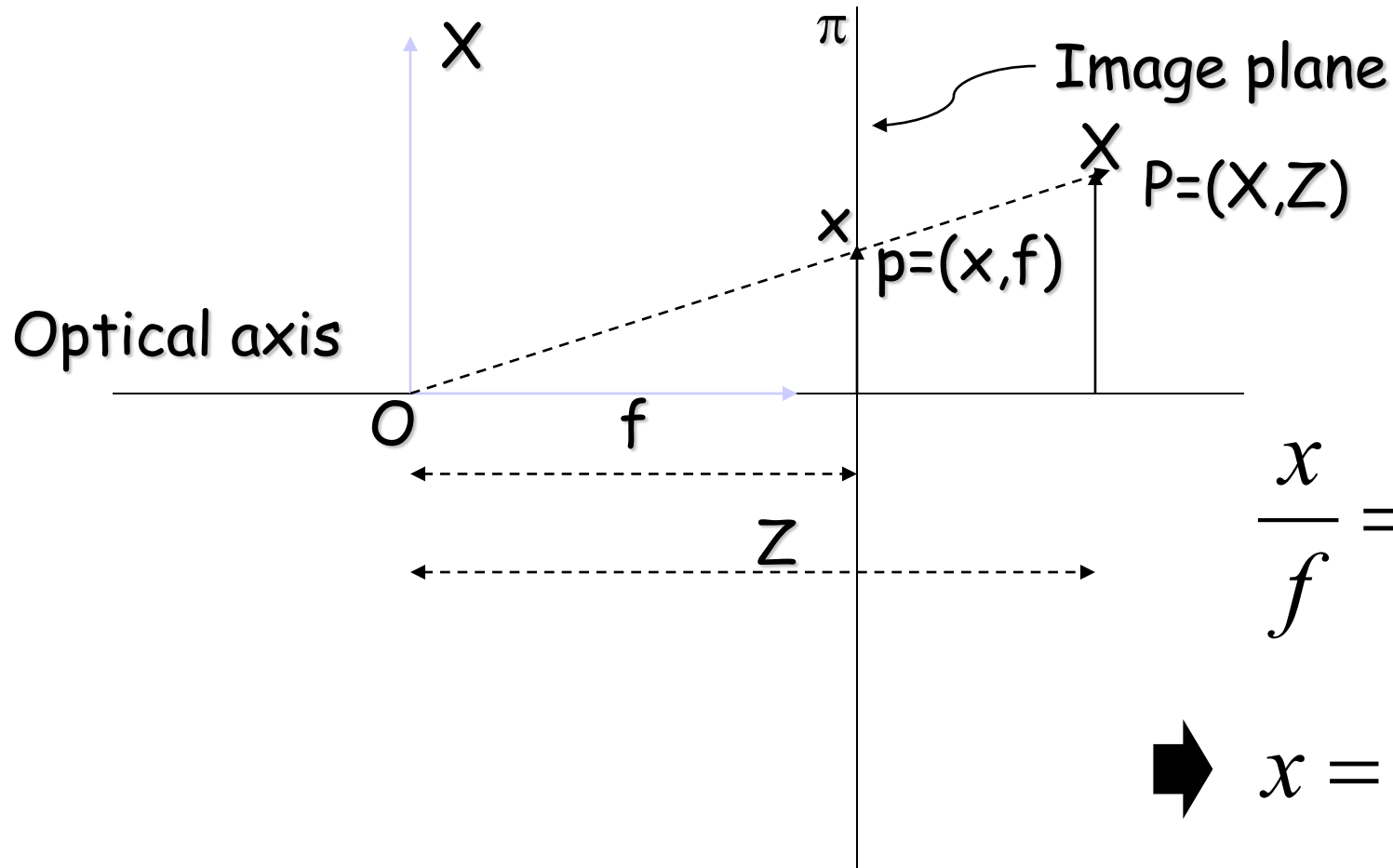
Non-linear

$$\begin{cases} x = f \frac{X}{Z} \\ y = f \frac{Y}{Z} \end{cases}$$

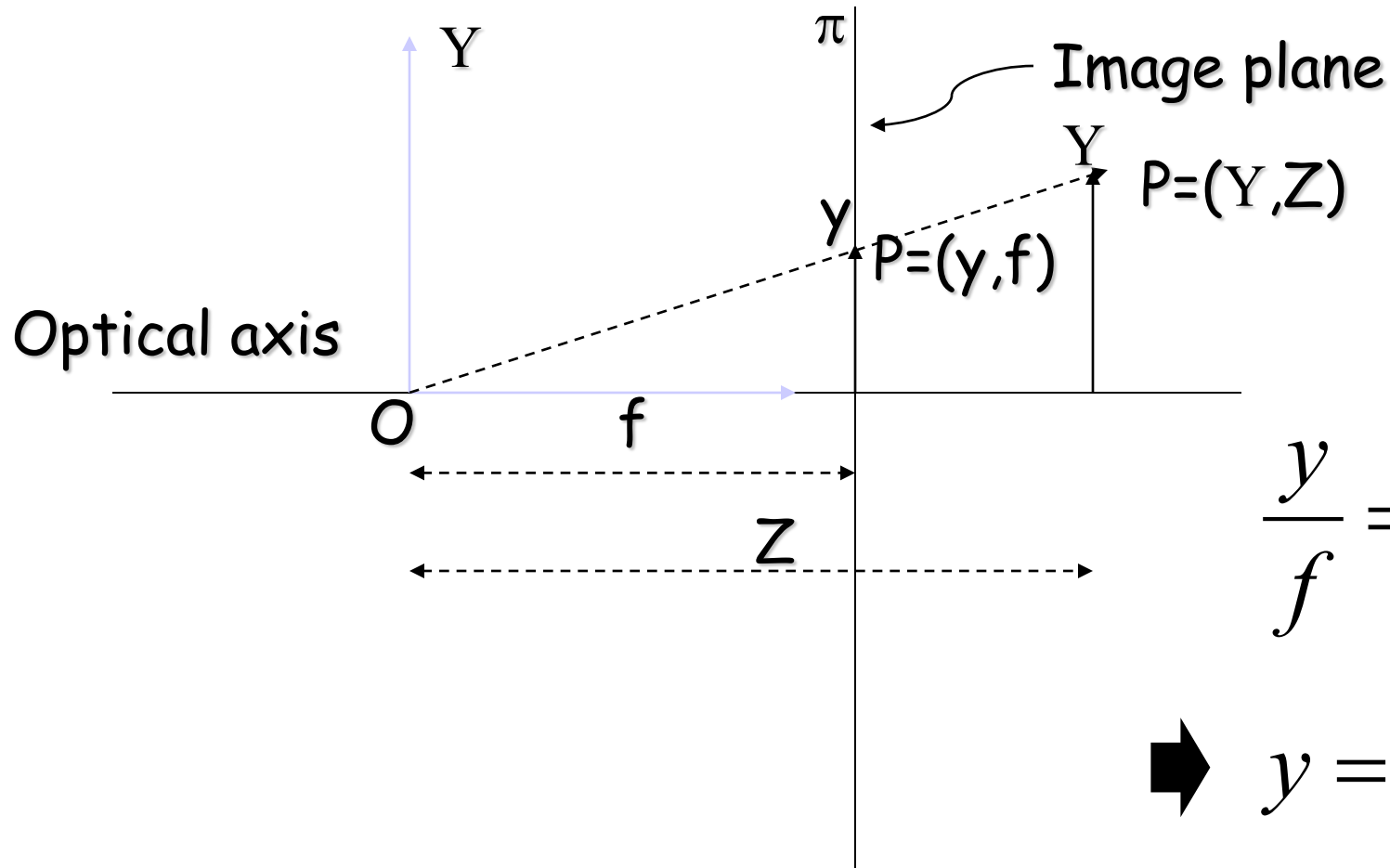


- Non-linear equations
- Any point on the ray OP has image p !!

2.3.1 3D Camera to 2D Image Coordinate: X-Z Plane



2.3.2 3D Camera to 2D Image Coordinate: Y-Z Plane



3.1.1 Homogeneous Matrix: 3D Camera to 2D Image Coordinate (1/2)

■ Using homogeneous coordinates:

$$\lambda \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{1} \end{bmatrix} = \begin{pmatrix} \alpha & \gamma & \mathbf{u}_0 \\ 0 & \beta & \mathbf{v}_0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{21} & \mathbf{R}_{31} & \mathbf{T}_1 \\ \mathbf{R}_{12} & \mathbf{R}_{22} & \mathbf{R}_{32} & \mathbf{T}_2 \\ \mathbf{R}_{13} & \mathbf{R}_{23} & \mathbf{R}_{33} & \mathbf{T}_3 \end{pmatrix} \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \\ \mathbf{Z} \\ 1 \end{bmatrix}$$

Scale Factor: λ ? Normalized to unit vector
 $[\mathbf{X}_c \ \mathbf{Y}_c \ \mathbf{Z}_c]$

Non-linear

$$\begin{cases} x = f \frac{X}{Z} \\ y = f \frac{Y}{Z} \end{cases}$$

Linear

1) Using homogeneous coordinates:

Intrinsic parameter

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Scaling factor ? at camera coordinate

2) Non-linear

$$x = \frac{x'}{w'} \quad y = \frac{y'}{w'}$$

$$p = M_{\text{int}} \cdot P_c$$

3.1.1 Homogenous Coordinates: $Ax=b$ (2/2)??

$$x' = z' \frac{x}{z}, \quad y' = z' \frac{y}{z}, \quad z' = z' = f$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{21} & R_{31} & T_1 \\ R_{12} & R_{22} & R_{32} & T_2 \\ R_{13} & R_{23} & R_{33} & T_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Scale Factor: λ ? Normalize to unit vector
 [Xc Yc Zc]

❑ Can we compute this in a 3x3 matrix?

➤ No, it's not linear => exists denominator

Why Homogenous Matrix
 ➔ Optimization Solution
 ➔ Solve the Loss Function

❑ Instead, use *homogeneous coordinates* for image plane coordinate

1) Linear: Homogenous matrix M , $b=Mx$

$$\begin{bmatrix} x'' \\ y'' \\ z'' \\ w'' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/z' & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

at camera coordinate

2) Non-linear equations

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x'' / w'' \\ y'' / w'' \\ z'' / w'' \end{bmatrix}$$

3.1.2 3D World to 3D Camera to 2D Image Coordinate

■ Using homogeneous coordinates:

□ **W to I:**

$$p(x,y) = M \cdot P_w(X_w,Y_w,Z_w)$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{21} & R_{31} & T_1 \\ R_{12} & R_{22} & R_{32} & T_2 \\ R_{13} & R_{23} & R_{33} & T_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Scale Factor: λ ? Normalize to unit vector

[Xc Yc Zc]

M is 3x4
M has 6 dof (3 R + 3 T)
(assuming f is known)

11 or 12 unknown para. of perspective project matrix, so it needs how many corresponding points btw 3D and 2D? At least 4 as a plane

□ Each image point $p(x, y)$ must satisfy:

1) Linear: Homogenous matrix M , $b=\dot{M}x$?

$$\begin{bmatrix} x' \\ y' \\ ?s' \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{31} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

s' as focal length f as scaling factor

2) Non-linear equations

$$\begin{cases} x = \frac{x'}{s'} = \frac{m_{11}X + m_{12}Y + m_{13}Z + m_{14}}{m_{31}X + m_{32}Y + m_{33}Z + m_{34}} \\ y = \frac{y'}{s'} = \frac{m_{21}X + m_{22}Y + m_{23}Z + m_{24}}{m_{31}X + m_{32}Y + m_{33}Z + m_{34}} \end{cases}$$

1

→ Linear Combination $y=w1_{\infty} * x1 + w2 * x2 + b =$

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$0 = m_{11}X + m_{12}Y + m_{13}Z + m_{14} - m_{31}Xx - m_{32}Yx - m_{33}Zx - m_{34}x$$

$$0 = m_{21}X + m_{22}Y + m_{23}Z + m_{24} - m_{31}Xy - m_{32}Yy - m_{33}Zy - m_{34}y$$

3.2.1 Optimization of Homogenous Coordinates ??

- Homogenous Matrix A , $b=Ax$

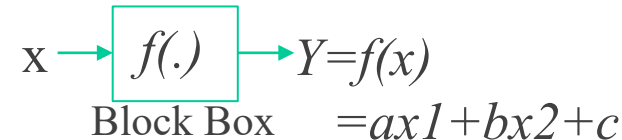
Linear Combination $y = w_1 * x_1 + w_2 * x_2 + b =$

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

- Basic function: cos or sin, (eigenvector)

→ orthonormal vector = orthogonal + normalization (unit vector)

- Parameter: $a, b, \dots$

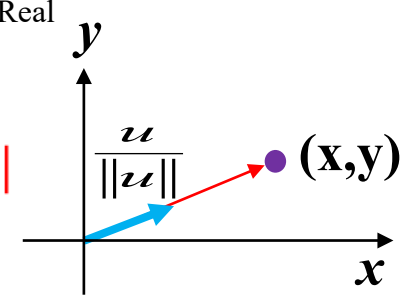


□ A simplistic view

- **Homogenous coordinates (matrix)** are a mechanism that allows us to associate **points and vectors** in space with vectors in $\mathbb{R}^{\text{Real}}$

Point or sample in high dimension:
20x20 pixels face image is a
point/sample/vector in 400 dimensions

❑ Location/Point u = Unit vector * Amplitude = $\frac{u}{||u||} * ||u||$
 = direction = scaling
 = rotation



□ **SVD: $(A-m) = UWV^T$; $(A-m)V = UW$; Gaussian Model $N(m, \sigma^2)$**
 where translation $\frac{\mu}{|\mu|}$

U : Orthonormal Matrix, Eigenvector, rotation

W : Scale Matrix, Eigenvalue, Standard Deviation

V: Rotation (Orthonormal) Matrix

3.2.2 Optimization of Homogenous Matrix: $Ax=b$

■ Solution/Optimization of Homogenous Matrix

-Local (linear) optimization (0 moment): 1.1 and 1.2 =>

-Global (non-linear) optimization: 2 => 2.1) LM 2.2) EM 2.3) SGD (Stochastic Gradient Decent by iterative process (Batch))

1.1 Closed-Form Solution:

Corresponding to

$Ax = 0 \Rightarrow A^T A = \text{Covariance Matrix} = \text{SVD} = U W U^T$: Smallest eigenvalue $> 0 \rightarrow$ eigenvector

1.2 Pseudo Inverse:

$$Ax = b, x = (A^T A)^{-1} * A^T b$$

2.0 Sum of Squared Difference: (max likelihood – exponential term)

Iteration

$$\min E = \sum [Ax - b]^2$$

- Why does deep learning apply Mean Squared Error
- $\min E = 1/B * \sum_1^B [Ax - b]^2$ where B is the batch size

2.1 $Ax = b'$: estimation value. b : ground truth, $E = \sum [b' - b]^2$

- Initial value estimation => Pseudo Inverse (linear approach)
- L-M (Levenberg-Marquardt Algorithm: non-linear approach)
 - First order Taylor series expansion

b.2 2nd order Taylor series expansion (sensitive to noise)

2.2 $Ax = b'$: estimation value. b'' : estimation value, $E = \sum [b' - b'']^2$

- Unsupervised Learning
- Use at Reinforcement learning for reward r

Machine learning
for prediction

a. EM (Expected-Maximization), initial $b' =$ average value

2.3 Stochastic Gradient Decent

- Deep Learning: Stochastic Gradient Decent

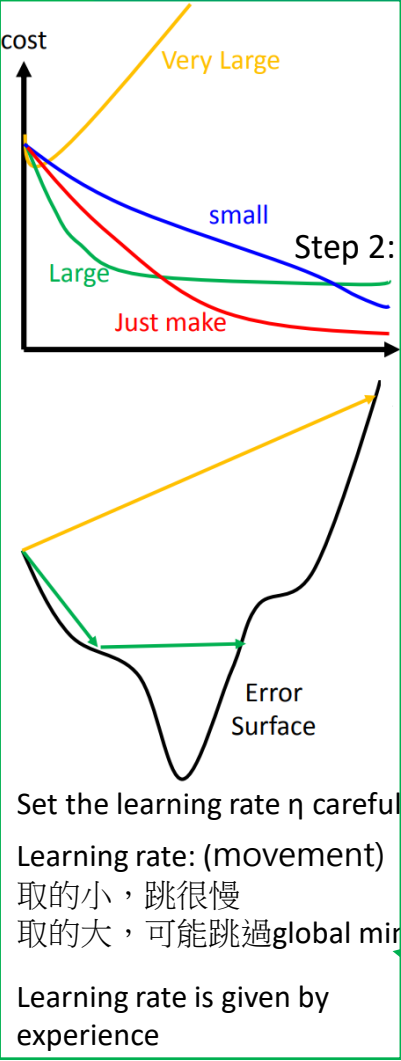
3. Lagrange Approach (outlier) with constraint

$$\min E = \sum [Ax - b]^2 + \lambda (x^2 + y^2)^2$$

- Constraint: GAN ...

3.2.2-1 Gradient Descent

Reference:[http://speech.ee.ntu.edu.tw/~tlkagk/courses/MLDS_2015_2/Lecture/DNN%20\(v4\).pdf](http://speech.ee.ntu.edu.tw/~tlkagk/courses/MLDS_2015_2/Lecture/DNN%20(v4).pdf)



Gradient Descent

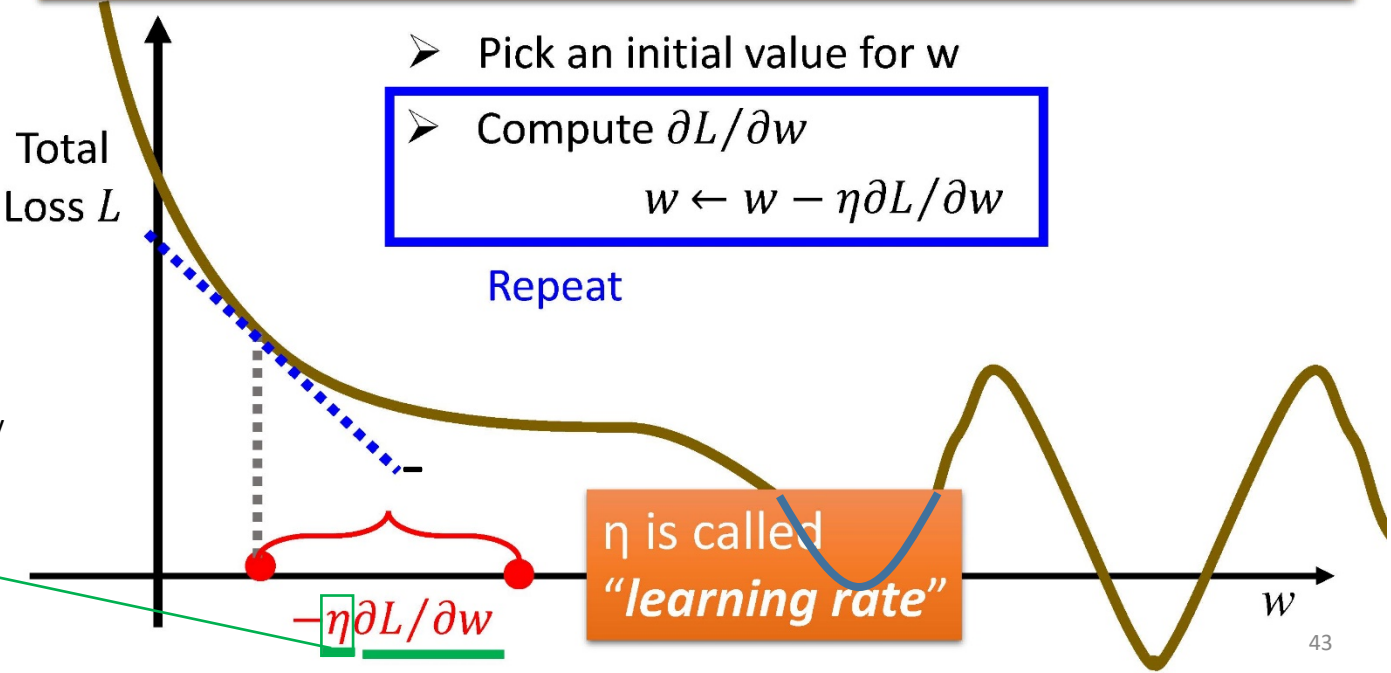
Network parameters $\theta = \{w_1, w_2, \dots, b_1, b_2, \dots\}$

Learning Target

Find the network parameters θ^* that minimize total loss

- Pick an initial value for w
 - Compute $\partial L / \partial w$
- $$w \leftarrow w - \eta \partial L / \partial w$$

Repeat



3.2.2-2 Lagrange multiplier method – Metrics

■ Using Ridge Regression to solve t:

Lambda called Lagrange Multiplier

岭回归(Ridge Regression)是在平方误差的基础上增加正则项

Loss $L = \sum_{i=1}^n \left(y_i - \sum_{j=0}^p w_j x_{ij} \right)^2 + \lambda \sum_{j=0}^p w_j^2$ With $\lambda=1000$

So this constraint term is for variance??

J: This is for bias optimization??

J: This is for variance optimization

通过确定λ的值可以使得在方差和偏差之间达到平衡：随着λ的增大，模型方差减小而偏差增大。

对w求导，结果为

variance

bias

$$\frac{\partial L}{\partial w} = 2X^T(Y - XW) - 2\lambda W$$

令其为0，可求得w的值：

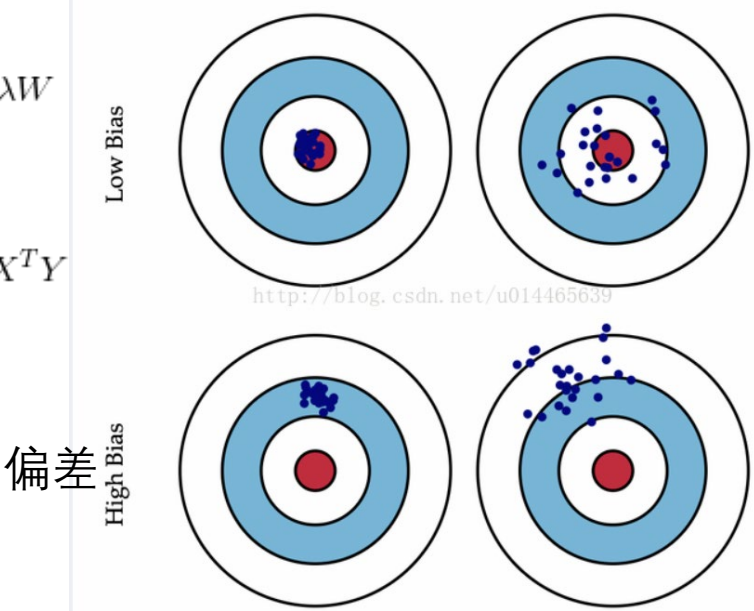
$$\hat{w} = (X^T X + \lambda I)^{-1} X^T Y$$

J: Enhance this term??

(平方差??)
方差

Low Variance

High Variance



Low variance is better

3.2.3 AI Bayesian Decision Rule – (1/3)

Maximum a Posteriori (MAP) Probability with Gaussian Model

1. Maximum a Posteriori (MAP) Probability with Gaussian Model

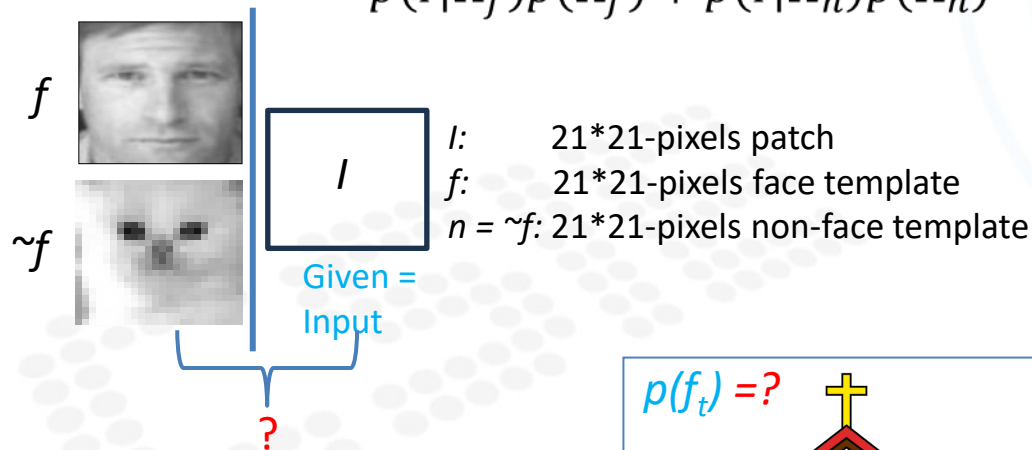
Decision
1) Posterior Prob. $\propto$ Likelihood Prob. * Priori Prob. Similarity Measure Imbalance, position

Prob. Of existing a face? MAP:

$$p(\Omega_f|I) = \frac{p(I|\Omega_f)p(\Omega_f)}{p(I|\Omega_f)p(\Omega_f) + p(I|\Omega_n)p(\Omega_n)}$$

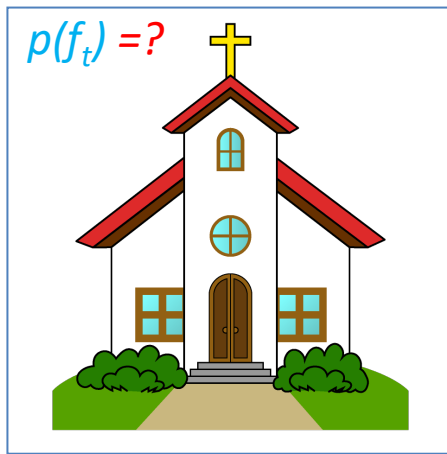
Max:

$$p(\Omega_n|I) = \frac{p(I|\Omega_n)p(\Omega_n)}{p(I|\Omega_f)p(\Omega_f) + p(I|\Omega_n)p(\Omega_n)}$$



3) Priori Prob. $p(f_t)$ or $p(!f) \neq 1/2$

> Example: position of face appearance probability at time t



2) Likelihood Prob. by Gaussian Model

> The similarity btw I and f

$$p(I|f) = \frac{1}{(2\pi)^{1/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2} (I - f) \Sigma^{-1} (I - f)\right]$$

Covariance matrix $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}$

4) Loss Function = $p(\Omega_f|I)$

= $p(f_t)$ * Similarity Measure Imbalance

- Similarity Measure:

$$\begin{aligned} \min SSD &= \min(I - f)^2 \\ &= \min(I^2 - 2If + f^2) \\ \min SSD &\rightarrow (I^2 + f^2) - 2(If) \end{aligned}$$

$$\therefore \max CC \rightarrow If$$

1,2,...,4,5,6,...8,9
 9,8,...,6,5,4,...2,1

Deep Learning:	min	max
1) MSE	SSD	correlation
2) Cross Entropy	9-1 = 8	9*1 = 9
	5-5 = 0	5*5 = 25

SSD: Sum of Square Difference

CC: Cross Correlation

B1-3 Homography Estimation $Ax=b$

Distance: norm

$r_{Mahalonobis} = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$

Mahalonobis distance

$d = \sqrt{(m_i - \hat{m}_i)^T (m_i - \hat{m}_i)}$

Euclidean distance = norm 1/2

1. Estimate Homography (H) Between the 3D Model Plane and its 2D Image:

H can be calculated by $\min_H \sum_i \|m_i - \hat{m}_i\|^2$ SSD: Sum of Squared Difference

- Assume 1: m_i is corrupted by Gaussian noise with mean 0 and covariance matrix Λ_{m_i} .

⇒ Probability Density Function (PDF) as likilhood probability:

$$P(\cdot) = \frac{1}{(2\pi)^{N/2} |\Lambda_{m_i}|^{1/2}} \exp \left(-\frac{1}{2} (m_i - \hat{m}_i)^T \Lambda_{m_i}^{-1} (m_i - \hat{m}_i) \right)$$

N: dimension of m_i , $N = 2$ in our case

⇒ Maximum likelihood estimation of H: $\min_H \sum_i (m_i - \hat{m}_i)^T \Lambda_{m_i}^{-1} (m_i - \hat{m}_i)$

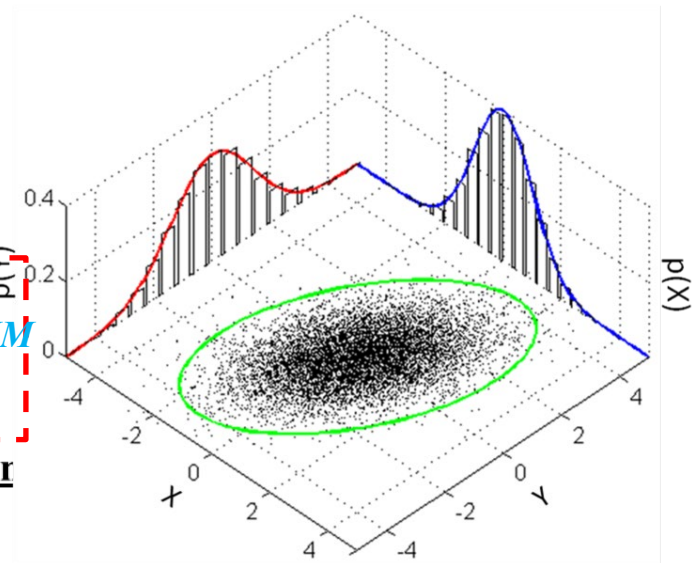
- Assume 2: for all i , $\Lambda_{m_i} = \sigma^2 I$.

$$\Rightarrow \min_H \sum_i (m_i - \hat{m}_i)^T \Lambda_{m_i}^{-1} (m_i - \hat{m}_i) = \min_H \sum_i \|m_i - \hat{m}_i\|^2$$

We can obtain H by applying Levenberg-Marqardt algorithm

- An initial guess using close form solution
- Levenberg-Marqardt algorithm

M_i : The 3D model points. (known)
 m_i : The 2D image points. (known)
 $\hat{m}_i$: Expected image points under the homography H.
 $\hat{m}_i = \begin{bmatrix} u \\ v \end{bmatrix} = \frac{1}{\bar{h}_3^T \hat{M}} \begin{bmatrix} \bar{h}_1^T \hat{M} \\ \bar{h}_2^T \hat{M} \end{bmatrix}$



3.2.3 Similarity Measure: Likelihood Probability [0.0, 1.0] (2/3)

2. Bayesian Decision Rule

$$x \in \begin{cases} \Omega_f \\ \Omega_n \end{cases} \begin{cases} \text{if } p(\Omega_f | x) \geq p(\Omega_n | x) \\ \text{otherwise} \\ \text{if } p(f|x) > \text{threshold} \end{cases} \Rightarrow \frac{p(x | \Omega_f)}{p(x | \Omega_n)} \begin{matrix} \text{face} \\ < \\ \text{nonface} \end{matrix} \frac{p(\Omega_n)}{p(\Omega_f)} = \gamma \quad \text{Constant by experiment}$$

1. Likelihood Probability [0.0,1.0]:

From sum-of-squared differences (SSD) to **Gaussian model (normal distribution) $N(\mu, \sigma^2)$**

$$P(x) = \frac{1}{\sqrt{(2\pi)^D |\Sigma|}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$$

$$P(I|x) = \frac{1}{\sqrt{(2\pi)^D |\Sigma|}} \exp\left(-\frac{1}{2} (x - I)^T \Sigma^{-1} (x - I)\right)$$

Covariance Matrix $\Sigma =$

$$\begin{bmatrix} (x_{11} - \mu)^2 & \dots & (x_{1D} - \mu)^2 \\ \vdots & \ddots & \vdots \\ (x_{D1} - \mu)^2 & \dots & (x_{DD} - \mu)^2 \end{bmatrix}$$

➔ The similarity btw input unknown patch I and ground truth face x
 I : 24x24-pixels patch; x : Face, $\sim x$: Non-Face

2. Likelihood Probability [0.0,1.0]:

From cross correlation (CC) to **normalized correlation coefficient (NCC) R**

$$|R(x, y)| = \left| \frac{\sum_{y'=0}^{h'-1} \sum_{x'=0}^{w'-1} [(T(x', y') - m^T) * I(x+x', y+y') - m^I]}{\sqrt{\sum_{y'=0}^{h'-1} \sum_{x'=0}^{w'-1} [(T(x', y') - m^T)]^2} \sqrt{\sum_{y'=0}^{h'-1} \sum_{x'=0}^{w'-1} [(I(x+x', y+y') - m^I)]^2}} \right|$$

3.2.3 Likelihood Probability by Gaussian Model (3/3-1/2)

2. Gaussian Distribution (Normal Distribution):

1) 當樣本為1-D (univariate)，高斯分佈遵守機率密度函數 (Probability Density Function - PDF):

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$x \in \{x_1, x_2, \dots, x_n\}$

$\mu = \frac{1}{n} \sum_{i=1}^n x_n$: mean (平均值 or 期望值): Average

$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$: standard deviation (標準差)

• Gaussian Distribution 物理意義:

• distance of x from μ

• degree of spread

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Annotations for the equation:

- Normalization term:** $\frac{1}{\sigma\sqrt{2\pi}}$ (between 0.0 ~ 1.0)
- Shift/Translation:** $(x-\mu)^2$
- Normalization/Scaling:** $2\sigma^2$
- Affine translation:** $-$ "Shift/translation, $- \sigma$ normalization/scaling"

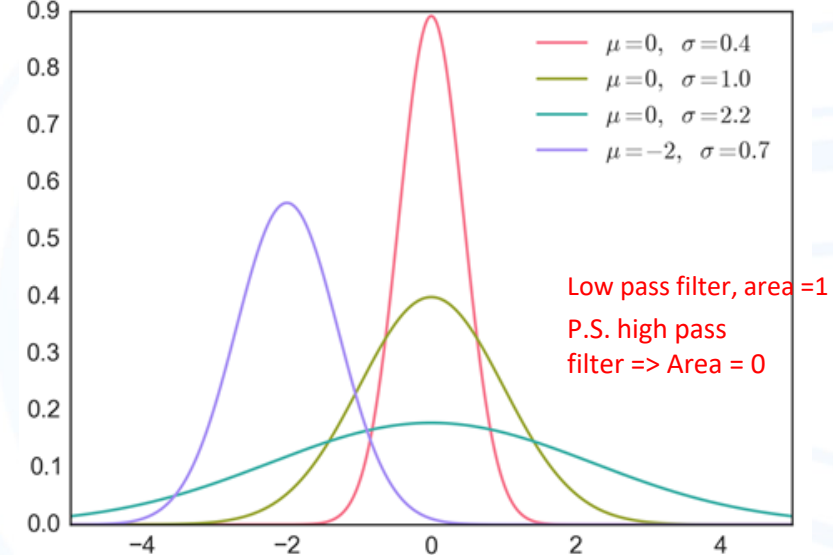
Normalization term:
between 0.0 ~ 1.0

- 位移: 回歸基準

- Exponential: 使大小值拉開

- 除標準差: 使 term 統一單位

不同平均數與標準差，所對應的機率密度函數曲線



- μ : Depend on Center-Position by μ (average),
- σ Depend on height and degree of spread by σ (standard deviation)

- a) Gaussian distribution (Probability)
 $m-\sigma$ (68%), $m+2\sigma$ (95%?), $m+3\sigma$ (99%)
- b) Gaussian Mixture Model (GMM)
➔ Weighted GMM estimated by EM

3.2.3 Likelihood Probability by Gaussian Model (3/3-2/2)

2) 當樣本為 n-D (multivariate), x 為 vector (向量化), 高斯分佈:

$$P(x) = \frac{1}{\sqrt{(2\pi)^D |\Sigma|}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

μ : 平均值 ($d \times 1$)

$$\Sigma = \begin{bmatrix} (x_{11} - \mu)^2 & \cdots & (x_{1D} - \mu)^2 \\ \vdots & \ddots & \vdots \\ (x_{D1} - \mu)^2 & \cdots & (x_{DD} - \mu)^2 \end{bmatrix}$$

D: 維度

$|\Sigma|$: 協方差的行列式

協方差矩陣(共變異數矩陣)

- covariance matrix

($d \times d$ symmetric matrix)

Note: 如果沒有 $-\mu$ 就是 correlation

tommy

• Exponential 的物理意義:

$$\exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

• If the variables are independent:

$$P(x) = \prod_i \frac{1}{\sigma_i \sqrt{2\pi}} \exp\left[-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right]$$

- SSD $\Sigma(y-y')^2$

y: Ground Truth

y': Estimation result

Cannot compare at

difference unit or space

Can compare at

difference unit or space

Euclidean distance

$d = \sqrt{(X - X')(X - X')^T}$

Mahalanobis distance

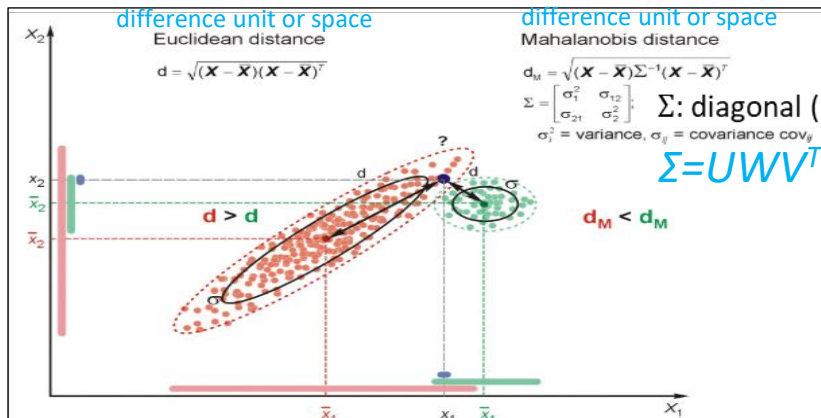
$d_M = \sqrt{(X - X')^T \Sigma^{-1} (X - X')}$

$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$

Σ : diagonal (只有對角線上有值)

σ_i^2 = variance, σ_{ij} = covariance cov_{ij}

$$\Sigma = U W U^T$$



'Euclidean Distance' VS 'Mahalanobis Distance'

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• Using Gaussian function At the Multivariate and Vector

Similarity measure $\rightarrow$ 1/Distance Measure

1) SSD / MSD problem $\Rightarrow$ Without Normalization Sigma

2) Mahalanobis Distance $\Rightarrow$ With Normalization Sigma + square root

$$\Sigma = U W U^T$$

• PCA (Principal Component Analysis)

- Gaussian model

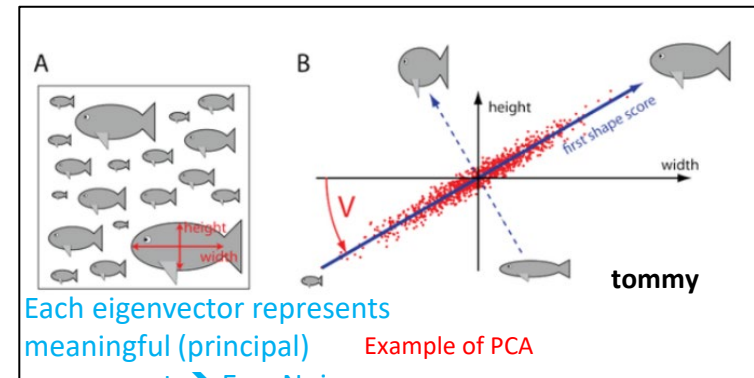
- because it is from Covariance matrix $N(u, \sigma^2)$

1) Covariance matrix (-u): Translation

2) Eigenvector U (& V): Direction = Rotation,

3) Eigenvalue W: Variance = Scaling

(present different covariance by PCA)



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Each eigenvector represents

meaningful (principal)

Example of PCA

component $\rightarrow$ Eye, Noise,

month...

Normalization to (0.0~1.0)

1) Gaussian

2) Softmax at DL

2.1) Range of element $\Rightarrow$ (0, 1)

2.2) Sum of elements $\Rightarrow$ 1

$$\sigma(\mathbf{z})_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}} \quad \text{for } j = 1, \dots, K.$$

Mahalanobias Distance

- Distance of Data Covariance
- The stronger the correlation between variables, the greater the value.

4.1 2D (Planar) Motions (Transformation) (1/4)

◆ 2D Planar Motion Transformations:

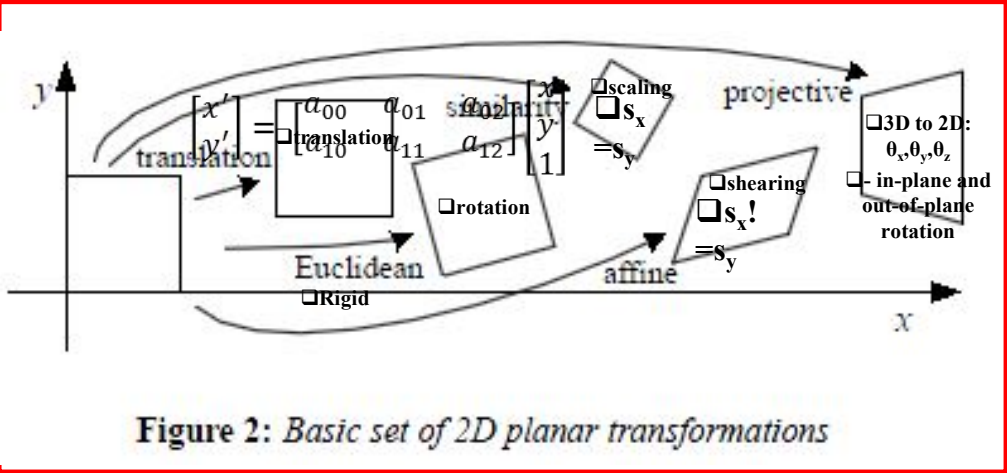


Figure 2: Basic set of 2D planar transformations

$$\begin{cases} x'=a_{00}x+a_{01}y+a_{02} \\ y'=a_{10}x+a_{11}y+a_{12} \end{cases}$$

I : Identity matrix (2x2).

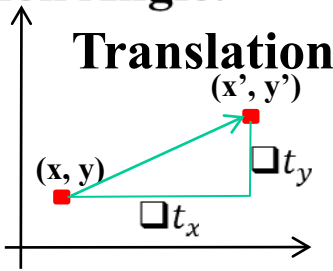
$\tilde{x} \ (\ x, y \ , \ 1)$: Homogeneous or projective 2D coordinate

$\tilde{x}' \ (x', y')$: Transformed coordinate

t (t_x, t_y) : Translation.

R : Rotation Matrix.

θ : Rotation Angle.



1) Translation Transformation: Translation

$$\tilde{x}' = \tilde{x} + t \quad \text{or} \quad \tilde{x}'_{2 \times 1} = [I_{2 \times 2} \ t_{2 \times 1}] \tilde{x}_{3 \times 1}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2) Euclidean Transformation (Rigid, 2D rigid body): Rotation + Translation

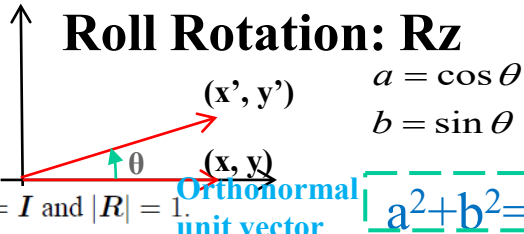
$$\tilde{x}' = R\tilde{x} + t \quad \text{or} \quad \tilde{x}' = [R \ t] \tilde{x}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Orthonormal unit vector

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

is an orthonormal rotation matrix with $RR^T = I$ and $|R| = 1$.



$$\begin{cases} a = \cos \theta \\ b = \sin \theta \end{cases}$$

$$a^2 + b^2 = 1$$

4.1 2D (Planar) Motions (Transformation) (2/4)

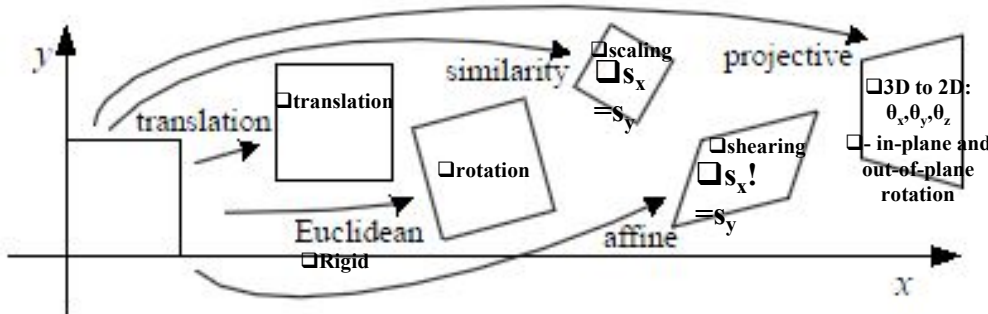


Figure 2: Basic set of 2D planar transformations

$\tilde{x} (x, y, 1)$: Homogeneous coordinates

$\tilde{x}' (x', y')$: Transformed coordinates.

$t (t_x, t_y)$: Translation.

R : Rotation matrix.

s : Scale factor.

θ : Rotation Angle.

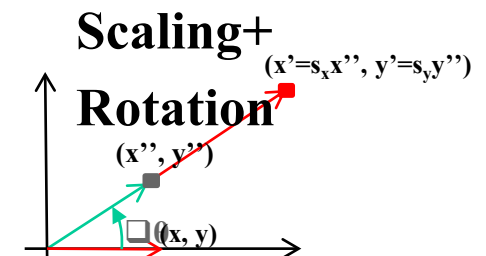
3) **Similarity Transformation** (Scaled rotation): Scaling ($s_x=s_y$) + Rotation + Translation

$$\tilde{x}' = sR\tilde{x} + t \quad \text{or} \quad \tilde{x}' = [sR \quad t]\tilde{x}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & -b & t_x \\ b & a & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad sR = \begin{bmatrix} s \cos \theta & -s \sin \theta \\ s \sin \theta & s \cos \theta \end{bmatrix}$$

$$a : s \cos \theta$$

$$b : s \sin \theta$$



where we no longer require that $a^2 + b^2 = 1$.

4.1 2D (Planar) Motions (Transformation) (3/4)

4) Affine Transformation: Scaling ($s_x \neq s_y$) + Shearing + Rotation + Translation any transformation that preserves parallel lines

$$\tilde{x}' = A\tilde{x} \quad \begin{cases} x' = a_{00}x + a_{01}y + a_{02} \\ y' = a_{10}x + a_{11}y + a_{12} \end{cases} \quad \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

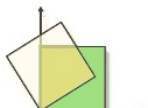
Translation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



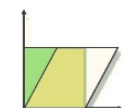
Rotation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



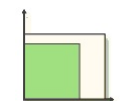
Shear:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & k_x \\ k_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



Scaling:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$\begin{bmatrix} x_1 \\ y_1 \\ w_1 \end{bmatrix} = \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} = 0 & h_{21} = 0 & h_{22} = 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Linear Combination:

$$x' = \frac{x_1}{w_1} = \frac{h_{00}x + h_{01}y + h_{02}}{h_{20}x + h_{21}y + h_{22}} = \frac{h'_{00}x + h'_{01}y + h'_{02}}{h'_{20}x + h'_{21}y + h'_{22}}$$

$$y' = \frac{y_1}{w_1} = \frac{h_{10}x + h_{11}y + h_{12}}{h_{20}x + h_{21}y + h_{22}} = \frac{h'_{10}x + h'_{11}y + h'_{12}}{h'_{20}x + h'_{21}y + h'_{22}}$$

1. Adding Translation ($\tilde{x}' = \tilde{x} + t$):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

2. Adding Rotation ($\tilde{x}' = R.\tilde{x} + t$):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

3.1. Adding Scaling ($\tilde{x}' = S.R.\tilde{x} + t$):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x \cos \theta & -S_x \sin \theta \\ S_y \sin \theta & S_y \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

3.2. Adding Shear ($\tilde{x}' = S.R.\tilde{x} + t$):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & k_x \\ k_y & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta + k_x \sin \theta & -\sin \theta + k_x \cos \theta \\ k_y \cos \theta + \sin \theta & -k_y \sin \theta + \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

4. All ($\tilde{x}' = S.S.R.\tilde{x} + t$):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} 1 & k_x \\ k_y & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x(\cos \theta + k_x \sin \theta) & S_x(-\sin \theta + k_x \cos \theta) \\ S_y(k_y \cos \theta + \sin \theta) & S_y(-k_y \sin \theta + \cos \theta) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Matrix form(2x3):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x(\cos \theta + k_x \sin \theta) & S_x(-\sin \theta + k_x \cos \theta) & t_x \\ S_y(k_y \cos \theta + \sin \theta) & S_y(-k_y \sin \theta + \cos \theta) & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$\tilde{x} (x, y, 1)$: Homogeneous coordinates.

$\tilde{x}' (x', y')$: Transformed coordinates.

$t (t_x, t_y)$: Translation.

R : Rotation matrix.

$$\tilde{x}' = A\tilde{x}$$

S_x : Scale factor to the x-axis.

S_y : Scale factor to the y-axis.

A : Affine Matrix.

θ : Rotation Angle.

k_x : Shear factor parallel to the x-axis.

k_y : Shear factor parallel to the y-axis.

H : Homogeneous matrix.

$$\begin{bmatrix} 1 & 1 \\ ky/kx & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \gamma & 1 \end{bmatrix}$$

?? Intrinsic skew factor??

Scale factor =?

Aspect Ratio = $\alpha/\beta = k_u/k_v$

4.1 2D (Planar) Motions (Transformation) (4/4)

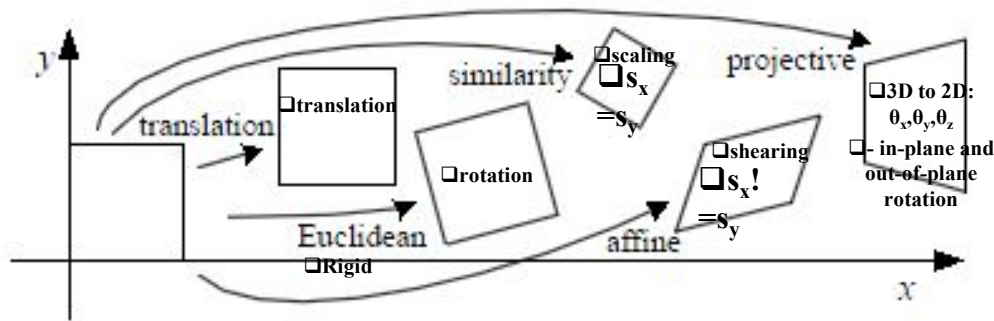


Figure 2: Basic set of 2D planar transformations

$\tilde{x} (x, y, 1)$: Homogeneous coordinates.

$\tilde{x}' (x', y')$: Transformed coordinates.

H : Homogeneous matrix.

Rotation:

In-Plane rotation + Out-of-plane rotation: $\theta_x, \theta_y, \theta_z$

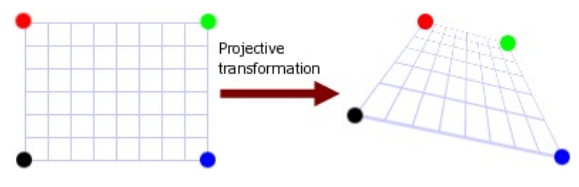
Including 3x3 intrinsic (or projection) parameters + affine transform with in-plane and out-of-plane rotations

5) Projective Transformation (Perspective Transform): 3D project to 2D

Projective transformations preserve **straight lines**.

$$\tilde{x}' \approx \tilde{H} \tilde{x}$$

$$\begin{bmatrix} x_1 \\ y_1 \\ w_1 \end{bmatrix} = \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \tilde{H} = \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix}$$



$$x' = \frac{x_1}{w_1} = \frac{h_{00}x + h_{01}y + h_{02}}{h_{20}x + h_{21}y + h_{22}} = \frac{h'_{00}x + h'_{01}y + h'_{02}}{h'_{20}x + h'_{21}y + 1}$$

$$y' = \frac{y_1}{w_1} = \frac{h_{10}x + h_{11}y + h_{12}}{h_{20}x + h_{21}y + h_{22}} = \frac{h'_{10}x + h'_{11}y + h'_{12}}{h'_{20}x + h'_{21}y + 1}$$

where $h'_{ij} = \frac{h_{ij}}{h_{22}}$

4.1 2D (Planar) Motions (Transformation)

Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\left[\begin{array}{c c} I & t \end{array} \right]_{2 \times 3}$	2	orientation + ...	
rigid (Euclidean)	$\left[\begin{array}{c c} R & t \end{array} \right]_{2 \times 3}$	3	lengths + ...	
similarity $S_x=S_y$	$\left[\begin{array}{c c} sR & t \end{array} \right]_{2 \times 3}$	4	angles + ...	
affine $S_x \neq S_y$	$\left[\begin{array}{c} A \end{array} \right]_{2 \times 3}$	6	parallelism + ...	
projective	$\left[\begin{array}{c} \tilde{H} \end{array} \right]_{3 \times 3}$	8	straight lines	

Table 1: *Hierarchy of 2D coordinate transformations. The 2×3 matrices are extended with a third $[0^T \ 1]$ row to form a full 3×3 matrix for homogeneous coordinate transformations.*

#D.O.F : Degrees Of Freedom

- 1) **Translation** : t_x, t_y
- 2) **Euclidean** : t_x, t_y, θ
- 3) **Similarity** : t_x, t_y, θ, s
- 4) **Affine** : $a_{00}, a_{01}, a_{02}, a_{10}, a_{11}, a_{12}$
- 5) **Projective** : $h'_{00}, h'_{01}, h'_{02}, h'_{10}, h'_{11}, h'_{12}, h'_{20}, h'_{21}$

6) **8-Parameter planar transformation:** $a0, a1, a2, a3, a4, a5, p0$ and $p1$

R. References

1. G. Bradski and A. Kaebler, *Learning OpenCV, Computer Vision with the OpenCV Library*, O'Reilly, 2008. ISBN-10: 0596516134 or ISBN-13: 978-0596516130.
2. Olivier Faugeras, *Three-Dimensional Computer Vision: A Geometric Viewpoint*, Third printing, 1999, The MIT Press.
3. R.C. Gonzalez and R.E. Woods, *Digital Image Processing*, 3rd edition, 2007.
4. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, Cambridge University Press, 2000.
5. B.K.P. Horn, *Robot Vision*, 1989, The MIT Press.
6. R. Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2010. ISBN-10: 1848829345 or ISBN-13: 978-1848829343.

❑ Keywords: CMOS Camera Vs. CCD Camera:

- <http://www.vvl.co.uk/whycmos/whitepaper.htm>
- <http://www.extremetech.com/article2/0,3973,75160,00.asp>
- <http://www.kodak.com/US/en/corp/researchDevelopment/technologyFeatures/cmos.shtml>
- <http://www.siliconimaging.com/>